

7

Angular momentum

In Chapter 4 we introduced the angular-momentum operators J_i as the generators of rotations. We showed that they form a pseudovector, so $J^2 = \sum_i J_i^2$ is a scalar. By considering the effect of rotations on vectors and scalars, we showed that the J_i commute with all scalar operators, including J^2 , and found that the commutator of J_i with a component of a vector operator is given by equation (4.31). From this result we deduced that the J_i do not commute with one another, but satisfy $[J_i, J_j] = i \sum_k \epsilon_{ijk} J_k$.

Although we have from the outset called the J_i 'angular-momentum operators', the only connection we have established between the J_i and angular momentum is tenuous and by no means justifies our terminology: we have simply shown that when the Hamiltonian is invariant under rotations about some axis $\hat{\alpha}$, and the system starts in an eigenstate of the corresponding angular-momentum operator $\hat{\alpha} \cdot \mathbf{J}$, it will subsequently remain in that eigenstate. Consequently, the corresponding eigenvalue is then a conserved quantity. In classical mechanics dynamical symmetry about some axis implies that the component of angular momentum about that axis is conserved, so it is plausible that the conserved eigenvalue is a measure of angular momentum. This suggestion will be substantiated in this chapter. Another important task for the chapter is to explain how the orientation of a system is encoded in the amplitudes for it to be found in different eigenstates of appropriate angular-momentum operators. We start by using the angular-momentum commutation relations to determine the spectrum of the J_i .

7.1 Eigenvalues of J_z and J^2

Since no two components of $\mathbf{J}$ commute, we cannot find a complete set of simultaneous eigenkets of two components of $\mathbf{J}$. We can, however, find a complete set of mutual eigenkets of J^2 and one component of $\mathbf{J}$ because $[J^2, J_i] = 0$. Without loss of generality we can orient our coordinates so that the chosen component of $\mathbf{J}$ is J_z . Let us label a ket which is simultaneously an eigenstate of J^2 and J_z as $|\beta, m\rangle$, where

$$J^2|\beta, m\rangle = \beta|\beta, m\rangle, \quad J_z|\beta, m\rangle = m|\beta, m\rangle. \quad (7.1)$$

We now define

$$J_{\pm} = J_x \pm iJ_y. \quad (7.2)$$

These objects clearly commute with J^2 , while their commutation relations with J_z are

$$\begin{aligned} [J_+, J_z] &= [J_x, J_z] + i[J_y, J_z] = -iJ_y - J_x = -J_+ \\ [J_-, J_z] &= [J_x, J_z] - i[J_y, J_z] = -iJ_y + J_x = J_-. \end{aligned} \quad (7.3)$$

Since $J_{\pm}$ commutes with J^2 , the kets $J_{\pm}|\beta, m\rangle$ are eigenkets of J^2 with eigenvalue β . Operating with J_z on these kets we find

$$\begin{aligned} J_z J_+ |\beta, m\rangle &= (J_+ J_z + [J_z, J_+]) |\beta, m\rangle = (m+1)J_+ |\beta, m\rangle \\ J_z J_- |\beta, m\rangle &= (J_- J_z + [J_z, J_-]) |\beta, m\rangle = (m-1)J_- |\beta, m\rangle. \end{aligned} \quad (7.4)$$

Thus, $J_+|\beta, m\rangle$ and $J_-|\beta, m\rangle$ are also members of the complete set of states that are eigenstates of both J^2 and J_z , but their eigenvalues with respect to J_z differ from that of $|\beta, m\rangle$ by ± 1 . Therefore we may write

$$J_{\pm}|\beta, m\rangle = \alpha_{\pm}|\beta, m \pm 1\rangle, \quad (7.5)$$

where $\alpha_{\pm}$ is a constant that we now evaluate. We do this by taking the length-squared of both sides of equation (7.5). Bearing in mind that $J_+^\dagger = J_-$, we find

$$\begin{aligned} |\alpha_+|^2 &= \langle \beta, m | J_- J_+ | \beta, m \rangle = \langle \beta, m | (J_x - iJ_y)(J_x + iJ_y) | \beta, m \rangle \\ &= \langle \beta, m | (J^2 - J_z^2 - J_z) | \beta, m \rangle = \beta - m(m+1). \end{aligned} \quad (7.6)$$

Similarly, $|\alpha_-|^2 = \beta - m(m-1)$, so

$$\alpha_{\pm} = \sqrt{\beta - m(m \pm 1)}. \quad (7.7)$$

The J_i are Hermitian operators, so $\langle \psi | J_i^2 | \psi \rangle = |J_i|\psi\rangle|^2 \geq 0$. Hence

$$\beta = \langle \beta, m | J^2 | \beta, m \rangle = \langle \beta, m | (J_x^2 + J_y^2 + J_z^2) | \beta, m \rangle \geq m^2. \quad (7.8)$$

So notwithstanding equation (7.5), it cannot be possible to create states with ever larger eigenvalues of J_z by repeated application of J_+ . All

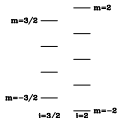


Figure 7.1 Going from $m_{\min}$ to $m_{\max}$ in an integer number of steps in the cases $j = \frac{1}{2}, 2$.

that can stop us doing this is the vanishing of α_+ when we reach some maximum eigenvalue $m_{\max}$ that from equation (7.7) satisfies

$$\beta - m_{\max}(m_{\max} + 1) = 0. \quad (7.9)$$

Similarly, α_- must vanish for a smallest value of m that satisfies

$$\beta - m_{\min}(m_{\min} - 1) = 0. \quad (7.10)$$

Eliminating β between (7.9) and (7.10) we obtain a relation between $m_{\max}$ and $m_{\min}$ that we can treat as a quadratic equation for $m_{\min}$. Solving this equation we find that

$$m_{\min} = \frac{1}{2}\{1 \pm (2m_{\max} + 1)\}. \quad (7.11)$$

The plus sign yields a value of $m_{\min}$ that is incompatible with our requirement that $m_{\min} \leq m_{\max}$, so we must have $m_{\min} = -m_{\max}$. To simplify the notation, we define $j = m_{\max}$, so that equation (7.9) becomes $\beta = j(j+1)$ and $-j \leq m \leq j$. Finally, we note that since an integer number of applications of J_- will take us from $|\beta, j\rangle$ through $|\beta, j-1\rangle, \dots$ to $|\beta, -j\rangle$, $2j$ must be an integer – see Figure 7.1. In summary, the eigenvalues of J^2 are $j(j+1)$ with $2j = 0, 1, 2, \dots$ and for each value of j the eigenvalues m of J_z are $(j, j-1, \dots, -j)$.

At this point we simplify the labelling of kets by defining $|j, m\rangle$ to be what has hitherto been denoted $|\beta, m\rangle$ with $\beta = j(j+1)$ – we clear a great deal of clutter from the page by replacing $|j(j+1), m\rangle$ with $|j, m\rangle$. The kets' eigenvalues with respect to J^2 are of course unaffected by this relabelling. Had we known at the outset that the eigenvalues of J^2 would be of the form $j(j+1)$, we would have used the new notation all along.

In summary, we can find simultaneous eigenstates of J^2 and one of the J_i , conventionally taken to be J_z . The eigenvalues of J^2 are $j(j+1)$ with $2j = 0, 1, \dots$, and for any given j the eigenvalues m of J_z then run from $+j$ to $-j$ in integer steps:

$$j \geq m \geq -j. \quad (7.12)$$

In order to move from the state $|j, m\rangle$ to the adjacent state $|j, m \pm 1\rangle$ we use the raising or lowering operators $J_{\pm}$ which act as

$$J_{\pm}|j, m\rangle = \alpha_{\pm}(m)|j, m \pm 1\rangle = \sqrt{j(j+1) - m(m \pm 1)}|j, m \pm 1\rangle. \quad (7.13)$$

These operators only change the J_z eigenvalue, so they just realign a given amount of total angular momentum, placing more (J_+) or less (J_-) along the z -axis. So far, we have not discovered how to alter the J^2 eigenvalue $j(j+1)$.

It is sometimes helpful to rewrite the constants $\alpha_{\pm}(m)$ of equation (7.13) in the form

$$\begin{aligned} \alpha_+(m) &= \sqrt{(j-m)(j+m+1)} \\ \alpha_-(m) &= \sqrt{(j+m)(j-m+1)}. \end{aligned} \quad (7.14)$$

These equations make it clear that the proportionality constants for different m satisfy

$$\begin{aligned} \alpha_+(m) &= \alpha_+(-m-1) & \alpha_+(m-1) &= \alpha_-(m) \\ \alpha_-(m) &= \alpha_-(-m+1) & \alpha_-(m) &= \alpha_+(-m). \end{aligned} \quad (7.15)$$

For example, when J_- lowers the highest state $|j, j\rangle$, we obtain the same proportionality constant as when J_+ raises the lowest state $|j, -j\rangle$; consequently, we only need to work out half the constants directly, because we can then infer the others.

By expressing $J_x = \frac{1}{2}(J_+ + J_-)$ in terms of the ladder operators, we observe that when we apply J_x to a state $|j, m\rangle$ for $j > 0$ we obtain a ket that differs from $|j, m\rangle$, so $|j, m\rangle$ is never an eigenket of J_x . Hence for $j > 0$ it is impossible to be certain what will be the result of measuring both J_z and J_x . It is trivial to see that this argument extends to the pair (J_z, J_y) , so for $j > 0$ it is impossible to be certain of the outcome of more than one of the J_i . If $j = 0$ the outcome measuring of any component of $\mathbf{J}$ is certainly 0, but a null vector has no direction. Consequently, there are no states in which the vector $\mathbf{J}$ has a well-defined direction. This situation contrasts with the case of the momentum vector $\mathbf{p}$, which can have a well-defined direction because its components commute with one another.

The angular-momentum vector is most nearly aligned with the z -axis when the body is in the state $|j, j\rangle$ and in this state the ratio of the squared angular momentum that lies in the xy plane to that parallel to the z -axis is

$$f = \frac{\langle j, j | (J_x^2 + J_y^2) | j, j \rangle}{\langle j, j | J_z^2 | j, j \rangle} = \frac{1}{j^2} \langle j, j | (J^2 - J_z^2) | j, j \rangle = \frac{1}{j}. \quad (7.16)$$

Since $\langle j, j | J_x | j, j \rangle = \langle j, j | J_y | j, j \rangle = 0$, we have no knowledge of the direction in which the xy component of angular momentum points. Macroscopic systems, which have huge values of j , have only a tiny fraction of their total angular momentum in the xy plane when they are in a state

$|j, j\rangle$, so the uncertain direction of this component of angular momentum does not lead to significant uncertainty in the direction of the total angular-momentum vector. By contrast, when $j = \frac{1}{2}$ there is twice as much angular momentum associated with the xy plane as with the z -axis, and the uncertain direction of this component makes it impossible to say more than that the angular-momentum vector lies somewhere in the northern rather than the southern hemisphere, or the western rather than the eastern hemisphere. Where it lies in the hemisphere is shrouded in quantum uncertainty. Even in the case $j = 1$, $m = 1$ there is as much angular momentum in the xy plane as parallel to the z -axis and the direction of the angular-momentum vector is very uncertain.

7.1.1 Rotation spectra of diatomic molecules

Knowledge of the spectrum of the angular-momentum operators enables us to understand an important part of the dynamics of a diatomic molecule such as carbon monoxide. For some purposes a CO molecule can be considered to consist of two point masses, the nuclei of the oxygen and carbon atoms, joined by a 'light rod' provided by the electrons. In this model the molecule's moment of inertia around the axis that joins the nuclei is negligible, while the same moment of inertia I applies to any perpendicular axis.

In classical mechanics the rotational energy of a rigid body is

$$E = \frac{1}{2} \left(\frac{J_x^2}{I_x} + \frac{J_y^2}{I_y} + \frac{J_z^2}{I_z} \right), \quad (7.17)$$

where the I_i are the moments of inertia about the body's three principal axes and $\mathbf{J}$ is the angular-momentum vector due to the body's spin. We conjecture that the equivalent formula links the Hamiltonian and the angular-momentum operators in quantum mechanics:

$$H = \frac{\hbar^2}{2} \left(\frac{J_x^2}{I_x} + \frac{J_y^2}{I_y} + \frac{J_z^2}{I_z} \right). \quad (7.18)$$

The best justification for adopting this formula is that it leads us to results that are confirmed by experiments.

In the case of an axisymmetric body, we orient our body such that the symmetry axis is parallel to the z -axis. Then $I = I_x = I_y$ and the Hamiltonian can be written

$$H = \frac{\hbar^2}{2} \left\{ \frac{J^2}{I} + J_z^2 \left(\frac{1}{I_z} - \frac{1}{I} \right) \right\}. \quad (7.19)$$

From this formula and our knowledge of the eigenvalues of J^2 and J_z , we can immediately write down the energies that form the spectrum of H :

$$E_{jm} = \frac{\hbar^2}{2} \left\{ \frac{j(j+1)}{I} + m^2 \left(\frac{1}{I_z} - \frac{1}{I} \right) \right\}, \quad (7.20)$$

where j is the total angular-momentum quantum number and $|m| < j$. In the case of a diatomic molecule such as CO, $I_z \ll I$ so the coefficient of m^2 is very much larger than the coefficient of $j(j+1)$ and states with $|m| > 0$ will occur only far above the ground state. Consequently, the states of interest have energies of the form

$$E_j = j(j+1) \frac{\hbar^2}{2I}. \quad (7.21)$$

For reasons that will emerge in §7.2.1, only integer values of j are allowed.

CO is a significantly dipolar molecule. The carbon atom has a smaller share of the binding electrons than the oxygen atom, with the result that it is positively charged and the oxygen atom is negatively charged. A rotating electric dipole would be expected to emit electromagnetic radiation. Because we are in the quantum regime, the radiation emerges as photons which, as we shall see, can add or carry away only one unit $\hbar$ of angular momentum. It follows that the energies of the photons that can be emitted or absorbed by a rotating dipolar molecule are

$$E_p = \pm (E_j - E_{j-1}) = \pm j \frac{\hbar^2}{I}. \quad (7.22)$$

Using the relation $E = h\nu$ between the energy of a photon and the frequency ν of its radiation, the frequencies in the rotation spectrum of the molecule are

$$\nu_j = j \frac{\hbar}{2\pi I}. \quad (7.23)$$

In the case of ^{12}CO , the coefficient of j evaluates to 113.1724 GHz and spectral lines occur at multiples of this frequency (Figure 7.2).

In the classical limit of large j , $\mathcal{J} = j\hbar$ is the molecule's angular momentum, and this is related to the angular frequency ω at which the molecule rotates by $\mathcal{J} = I\omega$. When in equation (7.23) we replace $j\hbar$ by $I\omega$, we discover that the frequency of the emitted radiation ν is simply the frequency $\omega/2\pi$ at which the molecule rotates around its axis. This conclusion makes perfect sense physically. Now, because of the form of the Hamiltonian, the energy eigenstates are also the eigenstates of J_z and J^2 . Therefore in any energy eigenstate, $\langle J^2 \rangle = j(j+1)$, and for low-lying states with $m = 0$ and $j \sim \mathcal{O}(1)$, $j(j+1)$ is significantly larger than j^2 . Therefore ν_j in (7.23) is smaller than the frequency at which the molecule rotates when it is in the upper state of the transition. On the other hand, ν_j is larger than the rotation frequency $\sqrt{(j-1)j} \frac{\hbar}{2\pi I}$ of the lower state. Hence the frequency at which radiation emerges lies between the rotation frequencies of the upper and lower states. Again this makes sense physically. As we approach the classical regime, j becomes large so $j(j+1) \simeq j^2 \simeq (j-1)j$ and the rotation frequencies of the upper and lower states converge, from above and below, on the frequency of the emitted radiation.

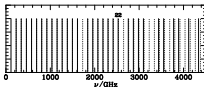


Figure 7.2 The rotation spectrum of CO. The full lines show the measured frequencies for transitions up to $j = 38 \rightarrow 37$, while the dotted lines show integer multiples of the lowest measured frequency. Up to the line for $j = 22 \rightarrow 21$ the dotted lines are obscured by the full lines except at one frequency for which measurements are not available. For $j \geq 22$ the separation between the dotted and full lines increases steadily as a consequence of the centrifugal stretching of the bond between the molecule's atoms. Measurements are lacking for several of the higher-frequency lines.

Measurements of radiation from 113 GHz and the first few multiples of this frequency provide one of the two most important probes of interstellar gas.¹ In denser, cooler regions, hydrogen atoms combine to form H_2 molecules, which are bisymmetric and do not have an electric dipole moment when they are simply rotating. Consequently, these molecules, which together with similarly uncommunicative helium atoms make up the great majority of the mass of cold interstellar gas, lack readily observable spectral lines. Hence astronomers are obliged to study the cold interstellar medium through the rotation spectrum of the few parts in 10^6 of CO that it contains.

Important information can be gleaned from the relative intensities of lines associated with different values of j in equation (7.23). The rate at which molecules emit radiation and thus the intensity of the line² is proportional to the number n_j of molecules in the upper state. In §6.4 we saw that when a gas is in thermal equilibrium at temperature T , the probability p_j that a given molecule is in a given state of energy E_j is proportional to the Boltzmann factor $\exp(-E_j/k_B T)$, where k_B is the Boltzmann constant (eq. 6.93a). From §7.1 we know that there are $2j + 1$ states with energy $E_j = j(j + 1)\hbar^2/2I$ because this is the number of possible orientations of the angular momentum for quantum number j . Hence the intensity of the line at frequency ν_j will be

$$I_j \propto (2j + 1) \exp(-E_j/k_B T) \quad (j > 0). \quad (7.24)$$

For $E_1 < k_B T$, I_j increases at small j before declining as the Boltzmann factor begins to overwhelm the degeneracy factor. Fitting this formula, which has only one free parameter (T), to observed line intensities enables one both to measure the temperature of the gas, and to check the correctness of the degeneracy factor.

¹ The other key probe is the hyperfine line of atomic hydrogen that will be discussed in Chapter 8.

² We neglect the absorption of photons after emission, which can actually be an important process, especially for ^{12}CO .

Figure 7.2 shows that for large values of the quantum number j , the spacing between lines in the spectrum diminishes in apparent violation of the prediction of equation (7.23). Lines with large j are generated by molecules that are spinning very rapidly. The bond between the nuclei is stretched like a spring by the centripetal acceleration of the nuclei. Stretching of the bond increases the moment of inertia I , and from equation (7.23) this decreases the frequency of the spectral lines (Problem 7.2).

7.2 Spin and orbital angular momentum

The Earth possesses two types of angular momentum: the **orbital angular momentum** associated with its near-circular orbit around the Sun, and its **spin angular momentum** associated with its spin around the axis that passes through the poles. Whereas orbital angular momentum relates to an external reference frame, in the Earth's case that defined by the location and rest frame of the Sun, spin is an intrinsic property of a body, and in this sense simpler and more fundamental.

Nearly all elementary particles (electrons, neutrinos, quarks, electrons, etc.) possess spin, so spin angular momentum plays a big role in quantum mechanics. Moreover, composite bodies such as atoms, protons, neutrons, pions, etc., frequently possess spin. A contribution to this spin may come from the orbital angular momentum of the component particles as they orbit around the body's barycentre, so orbital angular momentum may contribute to angular momentum that it is convenient to consider to be spin angular momentum. Nonetheless, it is conventional to use distinct symbols for orbital and spin angular momentum: the orbital angular-momentum operators are (L_x, L_y, L_z) and the spin angular-momentum operators are (S_x, S_y, S_z) . The symbols (J_x, J_y, J_z) are reserved for the operators related to the sum of orbital and spin angular momentum.

7.2.1 Orbital angular momentum

Let $\mathbf{x}$ and $\mathbf{p}$ be the position and momentum operators of the system. Then, inspired by classical mechanics, we define the dimensionless **orbital angular-momentum operators** by³

$$\mathbf{L} = \frac{1}{\hbar} \mathbf{x} \times \mathbf{p}, \quad \text{that is} \quad L_i = \frac{1}{\hbar} \sum_{jk} \epsilon_{ijk} x_j p_k. \quad (7.25)$$

From the rules of Table 2.1 and the Hermitian nature of $\mathbf{x}$ and $\mathbf{p}$, the Hermitian adjoint of L_i is

$$L_i^\dagger = \frac{1}{\hbar} \sum_{jk} \epsilon_{ijk} p_k^\dagger x_j^\dagger = \frac{1}{\hbar} \sum_{jk} \epsilon_{ijk} x_j p_k = L_i, \quad (7.26)$$

³ In many texts $\mathbf{L}$ is defined without the factor $\hbar^{-1}$. By making $\mathbf{L}$ dimensionless, this factor simplifies many subsequent formulae.

where we have used the fact that $[x_j, p_k] = 0$ for $j \neq k$. Thus the L_i are Hermitian and are likely to correspond to observables. We also define the **total orbital angular-momentum operator** by

$$L^2 = \mathbf{L} \cdot \mathbf{L} = L_x^2 + L_y^2 + L_z^2, \quad (7.27)$$

which is again Hermitian, and calculate a number of commutators. First, bearing in mind the canonical commutation relation (2.54), we have

$$\begin{aligned} [L_i, x_l] &= \frac{1}{\hbar} \sum_{jk} \epsilon_{ijk} [x_j p_k, x_l] = \frac{1}{\hbar} \sum_{jk} \epsilon_{ijk} x_j [p_k, x_l] = -i \sum_j \epsilon_{ijl} x_j \\ &= i \sum_j \epsilon_{ilj} x_j. \end{aligned} \quad (7.28)$$

Similarly

$$[L_i, p_l] = \frac{1}{\hbar} \sum_{jk} \epsilon_{ijk} [x_j p_k, p_l] = \frac{1}{\hbar} \sum_{jk} \epsilon_{ijk} [x_j, p_l] p_k = i \sum_j \epsilon_{ilj} p_j. \quad (7.29)$$

Notice that these commutation relations differ from the corresponding ones for J_i (eq. 4.30 and 4.32) only by the substitution of L for J . From these relations we can show that L_i commutes with the scalars $\mathbf{x}^2$, $\mathbf{p}^2$ and $\mathbf{x} \cdot \mathbf{p}$. For example

$$[L_i, p^2] = \sum_j [L_i, p_j^2] = i \sum_{jk} \epsilon_{ijk} (p_k p_j + p_j p_k) = 0, \quad (7.30)$$

where the last equality follows because the ϵ symbol is antisymmetric in jk while the bracket is symmetrical in these indices (see also Problem 7.3). We can now also calculate the commutator of one component of $\mathbf{L}$ with another. We have

$$[L_x, L_y] = \frac{1}{\hbar} [L_x, (z p_x - x p_z)] = i(-y p_x + x p_y) = i L_z. \quad (7.31)$$

Clearly each L_i commutes with itself, and the other non-zero commutators can be obtained from equation (7.31) by permuting indices. These commutators mirror the commutators (4.33) of the J_i , and from them it follows that $[L_i, L^2] = 0$.

$\mathbf{L}$ is a (pseudo) vector operator by virtue of the way it is constructed out of the vectors $\mathbf{x}$ and $\mathbf{p}$. It follows that L^2 is a scalar operator. Hence the way these operators commute with the total angular-momentum operators J_i follows from the work of §4.2:

$$[J_i, L_j] = i \sum_k \epsilon_{ijk} L_k \quad ; \quad [J_i, L^2] = 0. \quad (7.32)$$

Although p^2 and x^2 commute with L_i , the total angular-momentum operator J^2 does not:

$$[J^2, L_i] = \sum_j [J_j^2, L_i] = i \sum_{j,k} \epsilon_{jik} (L_k J_j + J_j L_k). \quad (7.33)$$

The right side vanishes only if $\mathbf{J} = \mathbf{L}$ because when these operators are not equal the final bracket is not symmetric in jk . The physical significance of $[J^2, L_i]$ being non-zero is that if our system is in a state of well-defined total angular momentum, in general there will be uncertainty in the amount of orbital angular momentum it has about any axis. We shall explore the consequences of this fact in §7.6.

L as the generator of circular translations In §4.1.1 we saw that when the system is displaced along a vector $\mathbf{a}$, its ket is transformed by the unitary operator $U(\mathbf{a}) = e^{-i\mathbf{a}\cdot\mathbf{p}/\hbar}$. We now imagine successively performing n translations through vectors $\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n\}$. Since each translation will cause $|\psi\rangle$ to be acted on by a unitary operator, the final state will be

$$\begin{aligned} U(\mathbf{a}_n) \dots U(\mathbf{a}_2)U(\mathbf{a}_1)|\psi\rangle &= \left\{ \prod_{i=1}^n \exp\left(\frac{i}{\hbar}\mathbf{a}_i \cdot \mathbf{p}\right) \right\} |\psi\rangle \\ &= \exp\left\{ \frac{i}{\hbar} \left(\sum_{i=1}^n \mathbf{a}_i \right) \cdot \mathbf{p} \right\} |\psi\rangle, \end{aligned} \quad (7.34)$$

where the second equality follows because the components of $\mathbf{p}$ commute with one another. Since the exponent in the last line is proportional to the overall displacement vector $\mathbf{A} = \sum_{i=1}^n \mathbf{a}_i$, the change in $|\psi\rangle$ is independent of the path that the system takes. In particular, if the path is closed, $\mathbf{A} = 0$ and $|\psi\rangle$ is unchanged.

Now consider the effect of moving the system in a circle centred on the origin and in the plane with unit normal $\mathbf{n}$. When we increment the rotation angle α by $\delta\alpha$, we move the system through

$$\delta\mathbf{a} = \delta\alpha \mathbf{n} \times \mathbf{x}. \quad (7.35)$$

The associated unitary operator is

$$\begin{aligned} U(\delta\mathbf{a}) &= \exp\left(-\frac{i}{\hbar}\delta\alpha (\mathbf{n} \times \mathbf{x}) \cdot \mathbf{p}\right) = \exp\left(-\frac{i}{\hbar}\delta\alpha \mathbf{n} \cdot (\mathbf{x} \times \mathbf{p})\right) \\ &= e^{-i\delta\alpha \mathbf{n} \cdot \mathbf{L}}. \end{aligned} \quad (7.36)$$

The unitary operator corresponding to rotation through a finite angle α is a high power of this operator. Since the exponent contains only one operator, $\mathbf{n} \cdot \mathbf{L}$, which inevitably commutes with itself, the product of the exponentials is simply

$$U(\alpha) = e^{-i\alpha \mathbf{n} \cdot \mathbf{L}}, \quad (7.37)$$

where $\alpha \equiv \alpha \mathbf{n}$.

The difference between the total and orbital angular-momentum operators is now apparent. When we rotate the system on a turntable through an angle α , the system's ket is updated by $e^{-i\alpha \mathbf{J}}$. When we move the system around a circle *without modifying its orientation*, the ket is updated by $e^{-i\alpha \mathbf{L}}$. The crucial insight is that the turntable both moves the system around a circle and reorients it (left panel Figure 7.3).



Figure 7.3 $\mathbf{J}$ both swings the particle around the origin and rotates its spin (left), while $\mathbf{L}$ moves the particle, but leaves the direction of the spin invariant (centre). $\mathbf{S}$ just rotates the spin (right).

The transformations of which $\mathbf{J}$ is the generator reflects both of these actions. The transformations of which $\mathbf{L}$ is the generator reflects only the translation.

Spectra of L^2 and L_z We have shown that the L_i commute with one another in exactly the same way that the J_i do. In §7.1 we found the possible eigenvalues of J^2 and J_z from the commutation relations and nothing else. Hence we can without further ado conclude that the possible eigenvalues of L^2 and L_z are $l(l+1)$ and m , respectively, with $-l \leq m \leq l$, where l is a member of the set $(0, \frac{1}{2}, 1, \frac{3}{2}, \dots)$.

In the last subsection we saw that $\mathbf{L}$ is the generator of translations on circles around the origin, and we demonstrated that when a complete rotation through 2π is made, the unitary operator that $\mathbf{L}$ generates is simply the identity. Consider the case in which we move the system right around the z -axis when it is in the eigenstate $|l, m\rangle$ of L^2 and L_z . The unitary operator is then $e^{-2\pi i L_z}$ and the transformed ket is

$$|l, m\rangle = e^{-2\pi i L_z} |l, m\rangle = e^{-2\pi i m} |l, m\rangle. \quad (7.38)$$

Since the exponential on the right side is equal to unity only for integer m , we conclude that L_z , unlike J_z , has only integer eigenvalues. Since for given l , m runs from $-l$ to l , it follows that l also takes only integer values. Thus the spectrum of L^2 is $l(l+1)$ with $l = 0, 1, 2, \dots$, and for given l the possible values of L_z are the integers in the range $(-l, l)$.

In §7.1.1 we saw that the emission spectrum of CO gas requires only transitions between states of integer j . This is because both the C and O atoms are spin-zero objects, so all the molecule's angular momentum is associated with the orbit of the reduced mass of the CO system about the molecule's centre of mass.

7.2.2 Spin angular momentum

The difference

$$\mathbf{S} = \mathbf{J} - \mathbf{L} \quad (7.39)$$

between the total angular momentum and the orbital angular momentum must be the spin angular momentum. We have found that $\mathbf{L}$ has exactly the same commutation relations as $\mathbf{J}$ with any function of the position

and momentum operators. From this fact and the definition (7.39), it follows that $\mathbf{S}$ commutes with all such functions. In particular $[\mathbf{S}, \mathbf{x}] = [\mathbf{S}, \mathbf{p}] = [\mathbf{S}, \mathbf{L}] = 0$. This confirms that $\mathbf{S}$ has nothing to do with a system's location or the way in which it may or may not be moving; $\mathbf{S}$ is associated with *intrinsic* properties of the system.

In the rest of this book we will make extensive use of commutation relations involving angular-momentum operators so Table 7.1 gathers them for later reference.

The states of atoms, nuclei and fundamental particles with which we have to deal are nearly always eigenstates of S^2 : the total amount of spin angular momentum is certain, the only uncertainty relates to the direction associated with this angular momentum. Given that $|\psi\rangle$ is an eigenket of S^2 , we can write

$$|\psi\rangle = \sum_{m=-s}^s \langle s, m | \psi \rangle |s, m\rangle. \quad (7.40)$$

The spin operators generate rotations of objects without moving them through space in any way: thus

$$|\psi'\rangle = U(\boldsymbol{\alpha})|\psi\rangle = e^{-i\boldsymbol{\alpha}\cdot\mathbf{S}}|\psi\rangle \quad (7.41)$$

is a state of the object that is identical to $|\psi\rangle$ in all respects except the object's orientation (§4.1.2). If $|\psi\rangle = |s, m\rangle$ is an eigenket of S^2 and S_z and $\boldsymbol{\alpha} = \alpha\mathbf{e}_z$ specifies a rotation around the z -axis, $U(\alpha)$ simply changes the phase of $|\psi\rangle$:

$$U(\alpha)|s, m\rangle = e^{-i\alpha S_z}|s, m\rangle = e^{-i\alpha m}|s, m\rangle. \quad (7.42)$$

Since s (and hence m) must be either an integer or a half-integer, after a rotation through 2π around the z -axis, we have either

$$|s, m\rangle \rightarrow |s, m\rangle \quad \text{for integer } m \quad (7.43a)$$

or

$$|s, m\rangle \rightarrow -|s, m\rangle \quad \text{for half-integer } m. \quad (7.43b)$$

Equation (7.43a) is as expected; under a 2π rotation, the object returns to its original state. However, equation (7.43b) says that an object with half-integer angular momentum does not return to its original state after a 2π rotation – the initial and final states are minus one another! This difference of behaviour between objects with integer and half-integer angular momentum is of fundamental importance, and determines many other characteristics of these objects. A result of quantum field theory is that fields with half-integer spin are always fermionic,⁴ while those with integer spin are bosonic. As we will see in §11.1, fermionic fields never

⁴ Relativity requires that the creation and annihilation operators of a field with half-integer spin satisfy the commutation relations of a Fermi oscillator (Problem 3.13), while those of a field with integer spin satisfy the commutation relations of the harmonic oscillator.

Table 7.1 Commutators involving angular momentum

$[J_i, v_j] = i \sum_k \epsilon_{ijk} v_k$	where $\mathbf{v}$ is any vector or pseudovector
$[J_i, s] = 0$	where s is any scalar or pseudoscalar
$[L_i, w_j] = i \sum_k \epsilon_{ijk} w_k$	where $\mathbf{w}$ is any vector or pseudovector function of only $\mathbf{x}$, $\mathbf{p}$ and constant scalars and pseudoscalars
$[L_i, f] = 0$	where f is any scalar or pseudoscalar function of only $\mathbf{x}$, $\mathbf{p}$ and constant scalars and pseudoscalars
$[S_i, w] = 0$	where w is any function of spatial operators

The following are important special cases of the above results

$$\begin{array}{lll}
 [J_i, J_j] = i \sum_k \epsilon_{ijk} J_k & [J_i, L_j] = i \sum_k \epsilon_{ijk} L_k & [J_i, S_j] = i \sum_k \epsilon_{ijk} S_k \\
 [L_i, L_j] = i \sum_k \epsilon_{ijk} L_k & [S_i, S_j] = i \sum_k \epsilon_{ijk} S_k & [L_i, S_j] = 0 \\
 [J_i, J^2] = 0 & [J_i, L^2] = 0 & [J_i, S^2] = 0 \\
 [L_i, L^2] = 0 & [L_i, S^2] = 0 & [S_i, L^2] = 0 \\
 [S_i, S^2] = 0 & [L^2, J^2] = 0 & [S^2, J^2] = 0
 \end{array}$$

Since $\mathbf{J} = \mathbf{L} + \mathbf{S}$ and therefore $J^2 = L^2 + S^2 + 2\mathbf{L} \cdot \mathbf{S}$ we also have

$$[L_i, J^2] = 2i \sum_{jk} \epsilon_{ijk} S_j L_k \qquad [S_i, J^2] = 2i \sum_{jk} \epsilon_{ijk} L_j S_k$$

attain macroscopic values: the quantum uncertainty in the value of a fermionic field is always on the same order as the value of the field itself. Bosonic fields, by contrast, can attain macroscopic values: values that are vastly greater than their quantum uncertainties. Consequently, classical physics – physics in the absence of quantum uncertainty – involves bosonic fields (the electromagnetic and gravitational fields are examples) but no fermionic field. Our intuition about what happens when an object is rotated has grown out of our experience of classical physics, so we consider that things return to their original state after rotation by 2π . If we had hands-on experience of fermionic objects, we would recognise that this is not generally true.

In the simplest case we have an object with $s = 0$, a **spin-zero** object, such as a Higgs boson. The sum in equation (7.40) contains only one eigenstate with $s = 0$, the state $|0, 0\rangle$. When we apply $U(\boldsymbol{\alpha})$ to a

state of a spin-zero object, the state emerges unchanged:

$$U(\boldsymbol{\alpha})|0, 0\rangle = e^{-i\boldsymbol{\alpha}\cdot\mathbf{S}}|0, 0\rangle = |0, 0\rangle. \quad (7.44)$$

Hence, a spin-zero object, like a perfect sphere, is completely unchanged by an arbitrary rotation. In view of their invariance under rotations, spin-zero particles are sometimes known as **scalar particles**.

Electrons, neutrinos, quarks, protons and neutrons are **spin-half** objects, so, for example the state of an electron can be written

$$|e^-\rangle = \sum_{m=-\frac{1}{2}}^{\frac{1}{2}} \langle \tfrac{1}{2}, m | e^-\rangle | \tfrac{1}{2}, m \rangle. \quad (7.45)$$

We saw at the end of §7.1 that the quantum uncertainty in the orientation of a spin-half object is very great.

Another important class consists of objects with $s = 1$, **spin-one** objects. The W and Z bosons fall in this class. For a spin-one object, the expansion (7.40) reduces to just $(2s + 1) = 3$ terms. For example, the state of a Z boson can be written

$$|Z\rangle = \sum_{m=-1}^1 \langle 1, m | Z \rangle |1, m\rangle. \quad (7.46)$$

The end of the angular-momentum vector of a spin-one object can be inferred to lie within a polar cap, or in the equatorial band that lies between opposite polar caps.

The larger a system's spin s , the more precisely we can locate the end of its angular-momentum vector. Amusingly, the same physical principle underlies the use of rifling in guns to stabilise the orientation of the projectile by imparting angular momentum to it as it flies down the barrel.

7.3 Physics of spin

The wavefunction of a particle with non-zero spin s has $2s + 1$ components $\psi_m(\mathbf{x})$, the m^{th} component giving the amplitude to be found at $\mathbf{x}$ in the spin state $|s, m\rangle$. The spin operators S_i act on these multi-component wavefunctions by matrix multiplication. We now derive the relevant matrices for the cases $s = \frac{1}{2}$ and $s = 1$.

7.3.1 Spin-half matrices

The state of any spin-half object may be expanded in terms of just two S_z eigenstates $|\frac{1}{2}, +\frac{1}{2}\rangle$ and $|\frac{1}{2}, -\frac{1}{2}\rangle$ which we will call $|+\rangle$ and $|-\rangle$ respectively. Equation (7.45) then reads $|\psi\rangle = a|+\rangle + b|-\rangle$. In this basis we can write the spin operators as (cf. equation 2.16)⁵

$$S_x = \begin{pmatrix} \langle +|S_x|+ \rangle & \langle +|S_x|-\rangle \\ \langle -|S_x|+ \rangle & \langle -|S_x|-\rangle \end{pmatrix}, \quad S_y = \begin{pmatrix} \langle +|S_y|+ \rangle & \langle +|S_y|-\rangle \\ \langle -|S_y|+ \rangle & \langle -|S_y|-\rangle \end{pmatrix}$$

$$S_z = \begin{pmatrix} \langle +|S_z|+ \rangle & \langle +|S_z|-\rangle \\ \langle -|S_z|+ \rangle & \langle -|S_z|-\rangle \end{pmatrix}. \quad (7.47)$$

The elements of the matrix S_z are trivially evaluated because $|\pm\rangle$ are the eigenkets of S_z with eigenvalues $\pm\frac{1}{2}$. To evaluate the other two matrices we notice that $S_x = \frac{1}{2}(S_+ + S_-)$, and $S_y = \frac{1}{2i}(S_+ - S_-)$, then use the relations $S_+|-\rangle = |+\rangle$ and $S_-|+\rangle = |-\rangle$ which follow from equation (7.13) with $j = \frac{1}{2}$. In this way we obtain

$$S_x = \frac{1}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}; \quad S_y = \frac{1}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}; \quad S_z = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (7.48)$$

The matrices appearing here are the **Pauli matrices**,

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}; \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}; \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad (7.49)$$

so we can write $\mathbf{S} = \frac{1}{2}\boldsymbol{\sigma}$. It is straightforward to verify that the square of any Pauli matrix is the identity matrix:

$$\sigma_i^2 = I. \quad (7.50)$$

This result implies that for any state $\langle S_x^2 \rangle = \langle S_y^2 \rangle = \langle S_z^2 \rangle = \frac{1}{4}$, which is consistent with the fact that the measurement of any component of $\mathbf{S}$ can produce only $\pm\frac{1}{2}$.

As an example of the application of spin-half matrices, let's find the coefficients a, b in the expansion

$$|+, \theta\rangle = a|+\rangle + b|-\rangle \quad (7.51)$$

of the state $|+, \theta\rangle$ that is the eigenket with eigenvalue $+\frac{1}{2}$ of $S_n = \mathbf{n} \cdot \mathbf{S}$ where $\mathbf{n} = (\sin\theta, 0, \cos\theta)$ is the unit vector that lies in the yz plane and is inclined at angle θ with respect to the z -axis. Physically $|+, \theta\rangle$ is the state in which the component of spin along $\mathbf{n}$ is certain to be $+\frac{1}{2}$. Adding $\sin\theta$ times the matrix for S_y to $\cos\theta$ times the matrix for S_z , the eigenvalue equation becomes

$$\frac{1}{2} \begin{pmatrix} \cos\theta & -i\sin\theta \\ i\sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \frac{1}{2} \begin{pmatrix} a \\ b \end{pmatrix}.$$

⁵ Here we are again slightly abusing the notation; S_i are taken to be both the spin operators and their matrix representations.

We have to solve this equation subject to the normalisation condition $|a|^2 + |b|^2 = 1$. From the first row of the matrix we deduce that

$$\frac{b}{a} = i \frac{1 - \cos \theta}{\sin \theta}. \quad (7.52)$$

From the trigonometric double-angle formulae we have $1 - \cos \theta = 2 \sin^2 \frac{1}{2} \theta$ and $\sin \theta = 2 \sin \frac{1}{2} \theta \cos \frac{1}{2} \theta$, so

$$\frac{b}{a} = i \frac{\sin \frac{1}{2} \theta}{\cos \frac{1}{2} \theta}. \quad (7.53)$$

The choices

$$\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} \cos \frac{1}{2} \theta \\ i \sin \frac{1}{2} \theta \end{pmatrix} \quad (7.54)$$

satisfy both equation (7.53) and the normalisation condition, so we have finally

$$|+, \theta\rangle = \cos \frac{1}{2} \theta |+\rangle + i \sin \frac{1}{2} \theta |-\rangle.$$

This derivation is readily generalised to give an expression for $|+, \mathbf{n}\rangle$, the eigenket of $\mathbf{n} \cdot \mathbf{S}$ for an arbitrary unit vector $\mathbf{n}$ (Problem 7.6).

7.3.2 Spin-one matrices

In the case that $s = 1$, three values of m are possible, $-1, 0, 1$, so the S_i may be represented by 3×3 matrices. The calculation of these matrices proceeds exactly as for spin-half, the main difference being that (7.13) now yields

$$\begin{aligned} S_+|-1\rangle &= \sqrt{2}|0\rangle, & S_+|0\rangle &= \sqrt{2}|1\rangle; \\ S_-|1\rangle &= \sqrt{2}|0\rangle, & S_-|0\rangle &= \sqrt{2}|-1\rangle. \end{aligned} \quad (7.55)$$

The result is

$$\begin{aligned} S_x &= \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, & S_y &= \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix} \\ S_z &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}. \end{aligned} \quad (7.56)$$

We use these matrices to find the state $|1, \theta\rangle$ in which a measurement of $\mathbf{n} \cdot \mathbf{S}$ will certainly yield one, where $\mathbf{n} = (0, \sin \theta, \cos \theta)$. With equations (7.56) the defining equation of this state can be written

$$\begin{pmatrix} \cos \theta & -\frac{1}{\sqrt{2}} \sin \theta & 0 \\ \frac{1}{\sqrt{2}} \sin \theta & 0 & -\frac{1}{\sqrt{2}} \sin \theta \\ 0 & \frac{1}{\sqrt{2}} \sin \theta & -\cos \theta \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}, \quad (7.57)$$

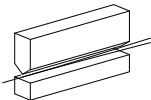


Figure 7.4 Schematic of a Stern-Gerlach filter. The atomic beam enters from the left. Between the pole pieces the magnetic field increases in intensity upwards, so atoms that have their spins aligned with $\mathbf{B}$ are deflected upwards and the other atoms are deflected downwards.

where $a \equiv \langle 1, z | 1, \theta \rangle$, $b \equiv \langle 0, z | 1, \theta \rangle$, and $c \equiv \langle -1, z | 1, \theta \rangle$. The first and third equations, respectively, yield

$$(\cos \theta - 1)a = \frac{i}{\sqrt{2}} \sin \theta b \quad \text{and} \quad \frac{i}{\sqrt{2}} \sin \theta b = (1 + \cos \theta)c. \quad (7.58)$$

Eliminating a and c in favour of b yields

$$\begin{pmatrix} a \\ b \\ c \end{pmatrix} = b \begin{pmatrix} \frac{1}{\sqrt{2}} \frac{\sin \theta}{\cos \theta - 1} \\ 1 \\ \frac{1}{\sqrt{2}} \frac{\sin \theta}{\cos \theta + 1} \end{pmatrix}. \quad (7.59)$$

The normalisation condition $|a|^2 + |b|^2 + |c|^2 = 1$ now implies that $b = \sin \theta / \sqrt{2}$ so

$$\begin{aligned} |1, \theta\rangle &= -i \frac{\sin^2 \theta}{2(1 - \cos \theta)} |1, 1\rangle + \frac{\sin \theta}{\sqrt{2}} |1, 0\rangle + i \frac{\sin^2 \theta}{2(1 + \cos \theta)} |1, -1\rangle \\ &= -i \frac{1}{2} (1 + \cos \theta) |1, 1\rangle + \frac{\sin \theta}{\sqrt{2}} |1, 0\rangle + i \frac{1}{2} (1 - \cos \theta) |1, -1\rangle. \end{aligned} \quad (7.60)$$

7.3.3 The Stern-Gerlach experiment

In 1922, Stern and Gerlach⁶ conducted some experiments with silver atoms that most beautifully illustrate the degree to which one can know the orientation of a spin-half object. In addition to this interest, these experiments provide clear examples of the standard procedure for extracting experimental predictions from the formalism of quantum mechanics.

A silver atom is an uncharged spin-half object and has a magnetic dipole moment $\boldsymbol{\mu}$ which can be used to track the atom's orientation. Classically, the potential energy of a magnetic dipole $\boldsymbol{\mu}$ in a magnetic field $\mathbf{B}$ is

$$H = -\boldsymbol{\mu} \cdot \mathbf{B}, \quad (7.61)$$

⁶ W. Gerlach and O. Stern, *Zeit. f. Physik*, **9**, 349 (1922).



Figure 7.5 Beam split by an SG filter and then the up-beam hits a second filter.

where the minus sign ensures that a dipole aligns with the field because this is its lowest-energy configuration. We align our coordinate system such that the z -axis lies along $\mathbf{B}$ and assume that the magnetic moment operator $\boldsymbol{\mu}$ is a constant 2μ times the spin operator $\mathbf{S}$. Then the Hamiltonian operator can be written

$$H = -2\mu B S_z. \quad (7.62)$$

The stationary states of this Hamiltonian are the eigenstates of S_z , which for a spin-half body such as a silver atom are the states $|\pm\rangle$ in which a measurement of S_z is certain to yield $\pm\frac{1}{2}$; the energies of these states are

$$E_{\pm} = \mp\mu B. \quad (7.63)$$

Hence the energy E_+ of the state $|+\rangle$ decreases with increasing B , while that of the state $|-\rangle$ increases with increasing B . Classically the force $\mathbf{F}$ on a particle is minus the particle's potential energy: $\mathbf{F} = -\nabla V$. So we expect a particle that's in the state $|+\rangle$ to be drawn into a region of high magnetic field strength and a particle in the state $|-\rangle$ to be repelled by such a region.

Stern and Gerlach exploited this effect to construct filters along the lines sketched in Figure 7.4. A powerful magnet has one pole sharpened to a knife edge while the other forms either a flat surface (as shown) or is slightly concave. With this geometry the magnetic field lines are close packed as they stream out of the knife edge, and then fan out as they approach the flat pole-piece. Consequently the intensity of the magnetic field increases towards the knife edge and the **Stern–Gerlach filter** sorts particles according to the orientation of their magnetic moments with respect to $\mathbf{B}$.

The experiments all start with a beam of silver atoms moving in *vacuo*, which is produced by allowing vapourised silver to escape from an oven through a suitable arrangement of holes – see Figure 7.5. When the beam passes into a filter, F1, it splits into just two beams of equal intensity. We explain this phenomenon by arguing that F1 ‘measures’ the component of $\mathbf{S}$ along the magnetic field, which we agree to be S_z . A measurement of S_z can yield only $\pm\frac{1}{2}$, so the splitting of the beam into two is explained. Given that there was nothing in the apparatus for producing the beam that favoured up over down as a direction for $\boldsymbol{\mu}$, it is to be expected that half of the atoms return $+\frac{1}{2}$ and half $-\frac{1}{2}$, so the two sub-beams have equal intensity. We block the sub-beam associated with $S_z = -\frac{1}{2}$ so that only particles with $S_z = \frac{1}{2}$ emerge from the filter.

We now place a second Stern–Gerlach filter, F2, in the path of the $|+\rangle$ sub-beam, as shown in Figure 7.5, and investigate the effect of rotating the

filter's magnetic axis $\mathbf{n}$ in the plane perpendicular to the incoming beam's direction. Let this be the yz plane. The incoming particles are definitely in the state⁷ $|+, z\rangle$ because they've just reported $+\frac{1}{2}$ on a measurement of S_z . F2 measures $\mathbf{n} \cdot \mathbf{S}$, where $\mathbf{n} = (0, \sin \theta, \cos \theta)$ with θ the direction between $\mathbf{n}$ and the z -axis. If $|+, \theta\rangle$ is the eigenket of $\mathbf{n} \cdot \mathbf{S}$ with eigenvalue $+\frac{1}{2}$, the amplitude that the measurement yields $+\frac{1}{2}$ is $\langle +, \theta | +, z \rangle$, which is just the complex conjugate of the coefficient of $|+\rangle$ in equation (7.55) for $|+, \theta\rangle$. Hence the probability that an atom will pass F2 is

$$P_2 = |\langle +, \theta | +, z \rangle|^2 = \cos^2 \frac{1}{2} \theta. \quad (7.64)$$

Thus, as θ is increased from 0 to π , the fraction of atoms that get through F2 declines from unity to zero, becoming $\frac{1}{2}$ when $\theta = \pi/2$ and the magnetic axes of F1 and F2 are at right angles. Physically it would be surprising if the fraction that passed F2 when $\theta = \pi/2$ were not a half since, when the magnetic moments of incoming atoms are perpendicular to the magnetic axis of a filter, there is nothing in the geometry of the experiment to favour the outgoing particles being parallel to the magnetic axis, rather than antiparallel. When $\theta = \pi$ the magnetic axes of the filters are antiparallel and it is obvious that every atom passed by F1 must be blocked by F2. The fact that θ has to be increased right up to π before all atoms are blocked by F2 is a consequence of the large quantum uncertainty in the orientation of a spin-half particle; if it is pointing somewhere in the upper z hemisphere, then there is some chance it is also pointing in any other hemisphere apart from the $-z$ one.

We now place a third filter, F3, in the atomic beam that emerges from F2. Let ϕ denote the angle between the magnetic axis of this filter and the z -axis. The atoms that emerge from F2 are in the state $|+, \theta\rangle$ because they've just returned $\frac{1}{2}$ on a measurement of $\mathbf{n} \cdot \mathbf{S}$, so the amplitude that these atoms get through F3 is $\langle +, \phi | +, \theta \rangle$. The amplitudes $\langle +, z | +, \phi \rangle$ and $\langle -, z | +, \phi \rangle$ can be obtained from equation (7.55) with ϕ substituted for θ . Hence

$$\begin{pmatrix} \langle +, z | +, \phi \rangle \\ \langle -, z | +, \phi \rangle \end{pmatrix} = \begin{pmatrix} \cos \frac{1}{2} \phi \\ i \sin \frac{1}{2} \phi \end{pmatrix}. \quad (7.65)$$

and the amplitude to pass F3 is

$$\begin{aligned} \langle +, \phi | +, \theta \rangle &= \sum_{s=\pm} \langle +, \phi | s, z \rangle \langle s, z | +, \theta \rangle \\ &= \left(\cos \frac{1}{2} \phi, -i \sin \frac{1}{2} \phi \right) \begin{pmatrix} \cos \frac{1}{2} \theta \\ i \sin \frac{1}{2} \theta \end{pmatrix} \\ &= \cos \frac{1}{2} (\phi - \theta). \end{aligned} \quad (7.66)$$

Thus the amplitude to pass F3 depends only on the angle $\phi - \theta$ between the magnetic axes of the filters, and the probability of passing F3 could

⁷ We relabel $|+\rangle \rightarrow |+, z\rangle$ to make clear that this is a state with spin up along the z -axis.

have been obtained simply by substituting this angle into equation (7.64). This conclusion is obvious physically, but it is satisfying to see it emerge automatically from the formalism.

An especially interesting case is when $\theta = \pi/2$ and $\phi = \pi$. In the absence of F2, F3 would now block every atom that passed F1. But with F2 present both F2 and F3 allow through half of the atoms that reach them, so a quarter of the atoms that leave F1 with $S_z = +\frac{1}{2}$ pass both filters. These atoms exit from F3 with $S_z = -\frac{1}{2}$. Introducing F2 changes the fraction of atoms that pass F3 because the measurement that F2 makes changes the states of the atoms. This is a central principle of quantum mechanics: no measurement can be made without slightly disturbing the system that is being measured, and if the system is small enough, the disturbance caused by a measurement can significantly affect the system's dynamics.

Stern–Gerlach experiment with spin-one atoms Consider the effect of using Stern–Gerlach filters on a beam of spin-one atoms. In the experiment depicted in Figure 7.5 each filter now splits the incoming beam into three sub-beams, and we block all but the beam associated with $m = +1$ along the magnetic axis. One third of the atoms that emerge from the collimating slits get through the first filter F1 because each value of m is equally probable at this stage.⁸ To calculate the fraction of atoms which then pass through F2, the magnetic axis of which is inclined at angle θ to that of F1, we need the amplitude $\langle 1, \theta | 1, z \rangle$. From equation (7.60) this is

$$\langle 1, \theta | 1, z \rangle = i\frac{1}{2}(1 + \cos \theta). \quad (7.67)$$

Hence the probability that an atom passes F2 after passing F1 falls from unity when $\theta = 0$ to zero when $\theta = \pi$ as we would expect. When $\theta = \pi/2$ the probability is $P_3 = \frac{1}{4}$, which is substantially smaller than the corresponding probability of $\frac{1}{2}$ found in (7.64) for the case of spin-half atoms.

From a classical point of view it is surprising that after F1 has selected atoms that have their angular momentum oriented parallel to the z -axis (in the sense that S_z takes the largest allowed value) there is a non-zero probability P_3 that the angular momentum is subsequently found to be, in the same sense, aligned with the y -axis. The explanation of this phenomenon is that, as we showed at the end of §7.1, even in the state $|1, z\rangle$ the object has as much angular momentum in the xy plane as along the z -axis. P_3 is the probability that this component is found to be parallel to the y -axis. Once the measurement of S_y has been made by F3, the atom is no longer in the state $|1, z\rangle$ and we are no longer certain to obtain 1 if we remeasure S_z .

⁸ The atoms emerge from the slits in an impure state (§6.3) and we set the probabilities for each value of m to $\frac{1}{3}$ in order to maximise the Shannon entropy of that state (§6.3.2). The equality of the probabilities is an instance of the general principle that every quantum state has equal a priori probability.

7.3.4 Precession in a magnetic field

A compass needle swings to the Earth's magnetic north pole because a magnetic dipole such as a compass needle experiences a torque when placed in a uniform magnetic field. Similarly, a proton that is in a magnetic field experiences a torque because it is a magnetic dipole. However, its response to this torque differs from that of a compass needle because it is a spinning body; instead of aligning with the magnetic field, it precesses around the field. This precession forms the basis for magnetic resonance imaging (MRI), which has become an enormously important diagnostic tool for chemistry and medicine. The theory of MRI is a fine example of the practical application of quantum mechanics in general and spin operators in particular.

In §7.3.3 we found that in a magnetic field $\mathbf{B}$ the stationary states of a spin-half magnetic dipole μ have energies

$$E_{\pm} = \mp \mu B. \quad (7.68)$$

The evolution in time of any spin state is

$$|\psi, t\rangle = a_- e^{-iE_- t/\hbar} |-\rangle + a_+ e^{-iE_+ t/\hbar} |+\rangle, \quad (7.69)$$

where $|\pm\rangle$ are the eigenkets of $\mathbf{B} \cdot \mathbf{S}$ and the constant amplitudes $a_{\pm}$ specify the initial condition $|\psi, 0\rangle = a_- |-\rangle + a_+ |+\rangle$.

Suppose that initially a measurement of the spin parallel to $\mathbf{n} = (\sin \theta, 0, \cos \theta)$ was certain to yield $\frac{1}{2}\hbar$. Then from Problem 7.6 we have that $a_- = \sin(\theta/2)$ and $a_+ = \cos(\theta/2)$. Hence at time t the proton's state is

$$\begin{aligned} |\psi, t\rangle &= \sin(\theta/2) e^{-iE_- t/\hbar} |-\rangle + \cos(\theta/2) e^{-iE_+ t/\hbar} |+\rangle \\ &= \sin(\theta/2) e^{i\phi/2} |-\rangle + \cos(\theta/2) e^{-i\phi/2} |+\rangle, \end{aligned} \quad (7.70a)$$

where

$$\phi(t) = \frac{2E_+ t}{\hbar} = \omega t \quad \text{where} \quad \omega = -\frac{2\mu_p B}{\hbar}, \quad (7.70b)$$

where μ_p is the magnetic moment of a proton. But from Problem 7.6 this is just the state $|+\rangle, \mathbf{n}'$ in which a measurement of the spin parallel to $\mathbf{n}' = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$ is certain to yield $\frac{1}{2}\hbar$. Consequently, the direction in which a measurement of the spin is certain to yield $\frac{1}{2}\hbar$ rotates around the direction of $\mathbf{B}$ at the frequency ω . This mirrors the behaviour expected in classical physics of a magnetic dipole of magnitude μ that has spin angular momentum $\frac{1}{2}\hbar$; its spin axis would precess around $\mathbf{B}$ at angular frequency ω (Problem 7.12).

When material that contains chemically bound hydrogen atoms is immersed in a powerful magnetic field, most of the protons align their spins with $\mathbf{B}$ in order to minimise their energy. Radiation of frequency ω has just the energy required to kick a proton into the higher-energy state in which its spin is anti-aligned with $\mathbf{B}$. Consequently, such radiation is readily absorbed by a sample, whereas radiation of neighbouring frequencies is not. As the analysis above shows, quantum interference between the aligned and anti-aligned states causes the expectation value of the

magnetic moment to precesses at angular frequency ω , and the precessing magnetic moment couples resonantly to the imposed radiation field.

The magnetic field at the location of a proton in a molecule has a contribution from the spins of the electrons that bind the proton, and this contribution varies slightly from one location to another. For example, in methanol, CH_3OH , the magnetic field experienced by the proton that is attached to the oxygen atom differs from those experienced by the protons that are attached to the carbon atom, and the proton that is on the other side of the carbon atom from the oxygen atom experiences a different field from the protons that are adjacent to the oxygen atom. Since the frequency ω of the resonant radiation is proportional to the magnitude of the magnetic field at the location of the proton, methanol has three different resonant frequencies for a given magnitude of the imposed magnetic field. Consequently, clues to the chemical structure of a substance can be obtained by determining the frequencies at which magnetic resonance occurs in a given imposed field.

7.3.5 The classical limit

An electric motor that is, say, 1 cm in diameter and weighs about 10 gm might spin at 100 revolutions per second. Its angular momentum would then be $\sim 10^{-3} \text{ kg m}^2 \text{ s}^{-1}$, which is $\sim 10^{31} \hbar$. Thus classical physics works with *extremely* large values of the integers s, m . It is interesting to understand how familiar phenomena emerge from the quantum formalism when s is no longer small.

For any value of s we can construct matrices that represent the angular-momentum operators. The matrix for S_z is diagonal with the eigenvalues $s, (s-1), \dots, -s$ down the diagonal. The matrices for S_x and S_y are evaluated in the usual way from S_+ and S_- and so are zero apart from strips one place above and below the diagonal. Using the relations (7.15) between the coefficients $\alpha_{\pm}(m)$ of the raising and lowering operators $S_{\pm}$ we then find

$$S_x = \frac{1}{2} \begin{pmatrix} 0 & \alpha(s-1) & 0 & \dots & \dots & 0 & 0 \\ \alpha(s-1) & 0 & \alpha(s-2) & & & 0 & 0 \\ 0 & \alpha(s-2) & 0 & \ddots & & \vdots & \vdots \\ \vdots & & \ddots & \ddots & \ddots & \vdots & \vdots \\ \vdots & \vdots & & \ddots & 0 & \alpha(s-2) & 0 \\ 0 & 0 & \dots & \alpha(s-2) & 0 & 0 & \alpha(s-1) \\ 0 & 0 & \dots & \dots & 0 & \alpha(s-1) & 0 \end{pmatrix}$$

$$= \frac{1}{2} [\alpha(m)\delta_{m,n-1} + \alpha(m-1)\delta_{m,n+1}] \quad (7.71a)$$

$$S_y = \frac{1}{2i} \begin{pmatrix} 0 & \alpha(s-1) & 0 & \dots & \dots & 0 & 0 \\ -\alpha(s-1) & 0 & \alpha(s-2) & \dots & \dots & 0 & 0 \\ 0 & -\alpha(s-2) & 0 & \ddots & \dots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \ddots & \ddots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \ddots & 0 & \alpha(s-2) & 0 \\ 0 & 0 & \dots & -\alpha(s-2) & 0 & 0 & \alpha(s-1) \\ 0 & 0 & \dots & \dots & 0 & -\alpha(s-1) & 0 \end{pmatrix}$$

$$= \frac{1}{2i} [-\alpha(m)\delta_{m,n-1} + \alpha(m-1)\delta_{m,n+1}] \quad (7.71b)$$

$$S_z = \begin{pmatrix} s & 0 & \dots & 0 & \dots & 0 & 0 \\ 0 & s-1 & & \vdots & & 0 & 0 \\ \vdots & & \ddots & & & \vdots & \vdots \\ 0 & & & m & & 0 & \\ \vdots & \vdots & & & \ddots & \vdots & \vdots \\ 0 & 0 & & \vdots & & 1-s & 0 \\ 0 & 0 & \dots & 0 & \dots & 0 & -s \end{pmatrix} \quad (7.71c)$$

$$= m \delta_{mn},$$

where the $\alpha(m)$ are what were called $\alpha_+(m)$ in (7.15), and the rows and columns of the matrix are labelled from $+s$ at the top left to $-s$ at the bottom right. In the same way as for spins $s = \frac{1}{2}$ and $s = 1$, it is straightforward (for a computer) to determine the amplitudes $a_m \equiv \langle s, m, z | s, s, \theta \rangle$ for measuring S_z to have value m , given that $\mathbf{n} \cdot \mathbf{S}$ certainly returns value s when $\mathbf{n} = (\sin \theta, 0, \cos \theta)$ is inclined at angle θ to the z -axis (Problem 7.13). The points in Figure 7.6 show the results of this calculation with $s = 100$ and three values of θ . The amplitudes peak around the values of the ordinate, $m = s \cos \theta$, that are marked with vertical lines. The larger the spin, the more sharply the amplitudes are peaked around these lines, so for the extremely large values of s that are characteristic of macroscopic systems, a_m is significantly non-zero only when m differs negligibly from $s \cos \theta$. Hence, in the classical limit the only values of S_z that have non-negligible probabilities of occurring lie in a narrow range around $s \cos \theta$, which is just the projection of the classical angular-momentum vector $\langle \mathbf{S} \rangle = s\mathbf{n}$ onto the z -axis. That is, in the classical limit the probability of measuring any individual value of S_z is small, but we are certain to find a value that lies close to the value predicted by classical physics.

The classical picture implies that when the angular-momentum vector is tipped in the xz plane at angle θ to the z -axis, the value of S_x should be $s \sin \theta$. We can show numerically that $\langle s, s, \theta | S_x | s, s, \theta \rangle = s \sin \theta$ by pre- and post-multiplying the matrix for S_x by the vector of amplitudes plotted in Figure 7.6 (Problem 7.13). However, this result is more easily obtained by writing $S_x = \cos \theta S_{x'} + \sin \theta S_{z'}$, where $S_{x'}$ and $S_{z'}$ are the spin operators associated with the coordinate system that has the z -axis parallel to the vector $\mathbf{n} = (\sin \theta, 0, \cos \theta)$ because $\langle s, s, \theta | S_{x'} | s, s, \theta \rangle = 0$ and $\langle s, s, \theta | S_{z'} | s, s, \theta \rangle = s$ so $\langle s, s, \theta | S_x | s, s, \theta \rangle = s \sin \theta$.

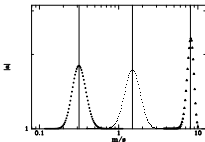


Figure 7.6 The points show the absolute values of the amplitudes $a_m \equiv \langle s, m, z | s, s, \theta \rangle$ for $s = 100$ and, from left to right, $\theta = 120^\circ, 80^\circ, 25^\circ$. For each value of θ , the vertical line shows the value of $\cos \theta$.

To determine the uncertainty in S_x we evaluate the expectation of S_x^2 . We have

$$\begin{aligned} \langle s, s, \theta | S_x^2 | s, s, \theta \rangle &= \frac{1}{4} \langle s, s, \theta | (S_{+'} + S_{-'}) (S_{+'} + S_{-'}) | s, s, \theta \rangle \\ &= \frac{1}{4} \langle s, s, \theta | S_{+'} S_{-'} | s, s, \theta \rangle = \frac{1}{2} s \end{aligned} \quad (7.72)$$

and

$$\begin{aligned} \langle s, s, \theta | S_x^2 | s, s, \theta \rangle &= \langle s, s, \theta | (\cos \theta S_{x'} + \sin \theta S_{z'})^2 | s, s, \theta \rangle \\ &= \langle s, s, \theta | (\cos^2 \theta S_{x'}^2 + \sin^2 \theta S_{z'}^2 \\ &\quad + \sin \theta \cos \theta \{ S_{x'} S_{z'} + S_{z'} S_{x'} \}) | s, s, \theta \rangle \\ &= \frac{1}{2} s \cos^2 \theta + s^2 \sin^2 \theta. \end{aligned} \quad (7.73)$$

The uncertainty in S_x , being $\sim (\langle S_x^2 \rangle - \langle S_x \rangle^2)^{1/2} = \sqrt{s/2} \cos \theta$, is therefore small compared with $\langle S_z \rangle$. A similar calculation shows that $\langle S_y \rangle$ vanishes and $\langle S_y^2 \rangle$ is small. Thus in the classical limit it is normal for all three components of $\mathbf{S}$ to have small uncertainties. However, S_x can be accurately determined precisely because there is *some* uncertainty in S_z : had we computed $\langle S_x \rangle$ and $\langle S_x^2 \rangle$ numerically, we would have found that quantum interference between states with different values of S_z is responsible for confining the likely values of S_x to a narrow range (Problem 7.13).

This is the third time we have found that the familiar world re-emerges through quantum interference between states in which some observable has well-defined values: in §2.3.3 we found that bullets can be assigned positions and momenta simultaneously through interference between states of well-defined momentum; in §3.2 we saw that an excited oscillator moves as a result of quantum interference between states of

well-defined energy; and now we find that a gyro has a well-defined orientation through quantum interference between states of well-defined angular momentum. In the classical regime a tiny fractional uncertainty in the value of an observable allows the vast numbers of states to have non-negligible amplitudes, and interference between these states narrowly defines the value of the variable that is canonically conjugate to the observable (§2.3.1).

7.4 Orbital angular-momentum eigenfunctions

We already know the possible eigenvalues of the operators L^2 and L_z . Now we find the corresponding eigenfunctions.

In the position representation, the L_i become differential operators. For example

$$L_z = \frac{1}{\hbar}(xp_y - yp_x) = -i \left(x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x} \right) \quad (7.74)$$

Let (r, θ, ϕ) be standard spherical polar coordinates. Then the chain rule states that

$$\frac{\partial}{\partial \phi} = \frac{\partial x}{\partial \phi} \frac{\partial}{\partial x} + \frac{\partial y}{\partial \phi} \frac{\partial}{\partial y} + \frac{\partial z}{\partial \phi} \frac{\partial}{\partial z}. \quad (7.75)$$

Using $x = r \sin \theta \cos \phi$, $y = r \sin \theta \sin \phi$ and $z = r \cos \theta$ we find

$$\frac{\partial}{\partial \phi} = r \sin \theta \left(-\sin \phi \frac{\partial}{\partial x} + \cos \phi \frac{\partial}{\partial y} \right) = \left(x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x} \right) = iL_z. \quad (7.76)$$

That is

$$L_z = -i \frac{\partial}{\partial \phi}. \quad (7.77)$$

Let $|l, m\rangle$ be a simultaneous eigenket of L^2 and m for the eigenvalues $l(l+1)$ and m , respectively. Then $L_z|l, m\rangle = m|l, m\rangle$ and the wavefunction $\psi_{lm}(\mathbf{x}) = \langle \mathbf{x} | l, m \rangle$ must satisfy the eigenvalue equation

$$-i \frac{\partial \psi_{lm}}{\partial \phi} = m \psi_{lm}. \quad (7.78)$$

The solution of this equation is

$$\psi_{lm}(r, \theta, \phi) = K_{lm}(r, \theta) e^{im\phi}, \quad (7.79)$$

where K_{lm} is an arbitrary function of r and θ . Since m is an integer, ψ_{lm} is a single-valued function of position.

In our determination of the spectra of J^2 and J_z in §7.1, important roles were played by the ladder operators $J_{\pm} = (J_x \pm iJ_y)$. If we define

$$L_{\pm} = L_x \pm iL_y, \quad (7.80)$$

then by analogy with equation (7.13) we will have that

$$L_{\pm}|l, m\rangle = \alpha_{\pm}|l, m \pm 1\rangle, \quad (7.81a)$$

where

$$\alpha_{\pm}(m) = \sqrt{l(l+1) - m(m \pm 1)}. \quad (7.81b)$$

It will be helpful to express $L_{\pm}$ in terms of partial derivatives with respect to spherical polar coordinates. We start by deriving a relation between partial derivatives that we will subsequently require. From the chain rule we have that

$$\frac{\partial}{\partial \theta} = r \cos \theta \left(\cos \phi \frac{\partial}{\partial x} + \sin \phi \frac{\partial}{\partial y} \right) - r \sin \theta \frac{\partial}{\partial z}. \quad (7.82a)$$

Multiplying the corresponding expression (7.76) for ϕ by $\cot \theta$ yields

$$\cot \theta \frac{\partial}{\partial \phi} = r \cos \theta \left(-\sin \phi \frac{\partial}{\partial x} + \cos \phi \frac{\partial}{\partial y} \right). \quad (7.82b)$$

Adding or subtracting i times (7.82b) to (7.82a) we obtain

$$\begin{aligned} \frac{\partial}{\partial \theta} \pm i \cot \theta \frac{\partial}{\partial \phi} &= r \cos \theta \left\{ (\cos \phi \mp i \sin \phi) \frac{\partial}{\partial x} + (\sin \phi \pm i \cos \phi) \frac{\partial}{\partial y} \right\} - r \sin \theta \frac{\partial}{\partial z} \\ &= r \cos \theta e^{\mp i \phi} \left(\frac{\partial}{\partial x} \pm i \frac{\partial}{\partial y} \right) - r \sin \theta \frac{\partial}{\partial z}. \end{aligned} \quad (7.83)$$

Multiplying through by $e^{\pm i \phi}$, we obtain the needed relation:

$$e^{\pm i \phi} \left(\frac{\partial}{\partial \theta} \pm i \cot \theta \frac{\partial}{\partial \phi} \right) = r \cos \theta \left(\frac{\partial}{\partial x} \pm i \frac{\partial}{\partial y} \right) - r \sin \theta e^{\pm i \phi} \frac{\partial}{\partial z}. \quad (7.84)$$

With this expression in hand we set to work on L_+ . In the position representation it is

$$\begin{aligned} L_+ &= -i \left(y \frac{\partial}{\partial z} - z \frac{\partial}{\partial y} \right) + \left(z \frac{\partial}{\partial x} - x \frac{\partial}{\partial z} \right) \\ &= z \left(\frac{\partial}{\partial x} + i \frac{\partial}{\partial y} \right) - (x + iy) \frac{\partial}{\partial z} \\ &= r \cos \theta \left(\frac{\partial}{\partial x} + i \frac{\partial}{\partial y} \right) - r \sin \theta e^{i \phi} \frac{\partial}{\partial z}, \end{aligned} \quad (7.85)$$

so with equation (7.84) we can write

$$L_+ = e^{i \phi} \left(\frac{\partial}{\partial \theta} + i \cot \theta \frac{\partial}{\partial \phi} \right). \quad (7.86a)$$

Similarly

$$\begin{aligned}
 L_- &= -i \left(y \frac{\partial}{\partial z} - z \frac{\partial}{\partial y} \right) - \left(z \frac{\partial}{\partial x} - x \frac{\partial}{\partial z} \right) \\
 &= -z \left(\frac{\partial}{\partial x} - i \frac{\partial}{\partial y} \right) + (x - iy) \frac{\partial}{\partial z} \\
 &= - \left\{ r \cos \theta \left(\frac{\partial}{\partial x} - i \frac{\partial}{\partial y} \right) - r \sin \theta e^{-i\phi} \frac{\partial}{\partial z} \right\} \\
 &= -e^{-i\phi} \left(\frac{\partial}{\partial \theta} - i \cot \theta \frac{\partial}{\partial \phi} \right).
 \end{aligned} \tag{7.86b}$$

The state $|l, l\rangle$ with the largest permissible value of m for given l must satisfy the equation $L_+|l, l\rangle = 0$. Using equations (7.79) and (7.86a), in the position representation this reads

$$\frac{\partial K_{ll}}{\partial \theta} - l \cot \theta K_{ll} = 0. \tag{7.87}$$

This is a first-order linear differential equation. Its integrating factor is $\exp(-l \int d\theta \cot \theta) = \sin^{-l} \theta$, so its solution is $K_{ll} = R(r) \sin^l \theta$, where R is an arbitrary function. Substituting this form of K_{ll} into equation (7.79), we conclude that

$$\psi_{ll}(r, \theta, \phi) = R(r) \sin^l \theta e^{il\phi}. \tag{7.88}$$

From equation (7.88) we can obtain the wavefunctions ψ_{lm} of states with smaller values of m simply by applying the differential operator L_- . For example

$$\begin{aligned}
 \psi_{l(l-1)}(r, \theta, \phi) &= \text{constant} \times R(r) e^{-i\phi} \left(-\frac{\partial}{\partial \theta} + i \cot \theta \frac{\partial}{\partial \phi} \right) \sin^l \theta e^{il\phi} \\
 &= \text{constant} \times R(r) \sin^{l-1} \theta \cos \theta e^{i(l-1)\phi}.
 \end{aligned} \tag{7.89}$$

Hence, the eigenfunctions of L^2 and L_z for given l all have the same radial dependence, $R(r)$. The function of θ, ϕ that multiplies R in ψ_{lm} is conventionally denoted Y_l^m and called a **spherical harmonic**. The normalisation of Y_l^m is chosen such that

$$\int d^2\Omega |Y_l^m|^2 = 1 \quad \text{with} \quad d^2\Omega = \sin \theta d\theta d\phi \tag{7.90}$$

the element of solid angle. We have shown that

$$Y_l^l \propto \sin^l \theta e^{il\phi} \quad \text{and} \quad Y_l^{l-1} \propto \sin^{l-1} \theta \cos \theta e^{i(l-1)\phi}. \tag{7.91}$$

The normalising constants can be determined by first evaluating the integral

$$\int d^2\Omega \sin^{2l} \theta = 4\pi 2^{2l} \frac{(l!)^2}{(2l+1)!} \tag{7.92}$$

Table 7.2 The first six spherical harmonics

	0	m ± 1	± 2
Y_0^m	$\sqrt{\frac{1}{4\pi}}$		
Y_1^m	$\sqrt{\frac{6}{8\pi}} \cos \theta$	$\mp \sqrt{\frac{3}{8\pi}} \sin \theta e^{\pm i\phi}$	
Y_2^m	$\sqrt{\frac{10}{32\pi}} (3 \cos^2 \theta - 1)$	$\mp \sqrt{\frac{15}{32\pi}} \sin 2\theta e^{\pm i\phi}$	$\sqrt{\frac{15}{32\pi}} \sin^2 \theta e^{\pm 2i\phi}$

involved in the normalisation of Y_l^m , and then dividing by the factor α_- of equation (7.81b) each time L_- is applied.

The spherical harmonics Y_l^m for $l \leq 2$ are listed in Table 7.2. Figures 7.7 and 7.8 show contour plots of several spherical harmonics. Since spherical harmonics are functions on the unit sphere, the figures show a series of balls with contours drawn on them. Since spherical harmonics are complex functions we had to decide whether to show the real part, the imaginary part, the modulus or the phase of the function. We decided it was most instructive to plot contours on which the real part is constant; when the real part is positive, the contour is full, and when it is negative, the contour is dotted.

For large l , Y_l^m is significantly non-zero only where $\sin \theta \simeq 1$, i.e., around the equator, $\theta = \pi/2$ – the leftmost panel of Figure 7.7 illustrates this case. The first l applications of L_- each introduce a term that contains one less power of $\sin \theta$ and an extra power of $\cos \theta$. Consequently, as m diminishes from l to zero, the region of the sphere in which Y_l^m is significantly non-zero gradually spreads from the equator toward the poles – compare the leftmost and rightmost panels of Figure 7.7. These facts make good sense physically: Y_l^m is the wavefunction of a particle that has essentially all its orbital angular momentum parallel to the z -axis, so the particle should not stray far from the xy plane. Hence Y_l^l , the amplitude to find the particle at θ , should be small for θ significantly different from $\pi/2$. As m diminishes the orbital plane is becoming more inclined to the xy plane, so we are likely to find the particle further and further from the plane. This is why Y_l^m increases away from the equator as m decreases.

For large l the phase of Y_l^m changes rapidly with ϕ (leftmost panel of Figure 7.7). This is to be expected, because the particle's large orbital angular momentum, lh , implies that the particle has a substantial tangential motion within the xy plane. From classical physics we estimate its tangential momentum at $p = lh/r$, and from quantum mechanics we know that this implies that the wavefunction must change its phase at a rate $p/\hbar = l/r$ radians per unit distance. This estimate agrees precisely with the rate of change of phase with distance around the equator arising from the factor $e^{i\phi}$ in Y_l^m . When m is significantly smaller than l (rightmost panel of Figure 7.7), the rate of change of the wavefunction's phase with increasing ϕ is smaller because the particle's tangential momentum is not all in the direction of increasing ϕ . Hence $Y_l^m \propto e^{im\phi}$.

For any value of m , L_x and L_y both have zero expectation values, as follows immediately from the relation $L_x = \frac{1}{2}(L_+ + L_-)$. So the orienta-

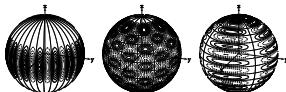


Figure 7.7 Contours of $\Re(Y_{15}^m)$ on the unit sphere for $m = 15$ (left), $m = 7$ (centre) and $m = 2$ (right). The contours on which $\Re(Y_{15}^m) = 0$ are the heavy curves, while contours on which $\Re(Y_{15}^m) < 0$ are dotted. Contours of the imaginary part of Y_{15}^m would look the same except shifted in azimuth by half the distance between the heavy curves of constant azimuth.

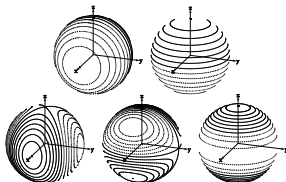


Figure 7.8 Top row: contours of $\Re(Y_1^m)$ for $m = 1$ (left) and 0 (right) with line styles having the same meaning as in Figure 7.7. Contours of the imaginary part of Y_1^m would look the same as the left panel but with the circles centred on the y -axis. Bottom row: contours of $\Re(Y_2^m)$ for $m = 2$ (left), $m = 1$ (centre) and $m = 0$ (right).

tion of the component of the angular-momentum vector that lies in the xy plane is completely uncertain. Because of this uncertainty, the modulus of Y_l^m is independent of ϕ , so there is no trace of an inclined orbital plane when $m < l$. An orbital plane becomes defined if there is some uncertainty in L_z , with the result that there are non-zero amplitudes $\psi_m = \langle l, m | \psi \rangle$ for several values of m . In this case quantum interference between states of well-defined L_z can generate a peak in $|\langle \mathbf{x} | \psi \rangle|^2$ along a great circle that is inclined to the equator.

7.4.1 Orbital angular momentum and parity

In §4.1.4 we defined the parity operator P , which turns a state with wavefunction $\psi(\mathbf{x})$ into the state that has wavefunction $\psi'(\mathbf{x}) \equiv \psi(-\mathbf{x})$. We now show that wavefunctions that are proportional to a spherical harmonic Y_l^m are eigenfunctions of P with eigenvalue $(-1)^l$.

In polar coordinates the transformation $\mathbf{x} \rightarrow -\mathbf{x}$ is effected by $\theta \rightarrow \pi - \theta$, $\phi \rightarrow \phi + \pi$. Under this mapping, $\sin \theta = \sin(\pi - \theta)$ is unchanged, while $e^{i\phi} \rightarrow e^{i\pi} e^{i\phi} = (-1)^l e^{i\phi}$. By equation (7.91), $Y_l^m \propto \sin^l \theta e^{im\phi}$, so $Y_l^m \rightarrow (-1)^l Y_l^m$. That is, Y_l^m has even parity if l is an even number and odd parity otherwise.

In §4.1.4 we saw that $\mathbf{x}$ and $\mathbf{p}$ are odd-parity operators: $P\mathbf{x} = -\mathbf{x}P$. From this and the fact that the orbital angular-momentum operators L_i are sums of products of a component of $\mathbf{x}$ and a component of $\mathbf{p}$, it follows that both the L_i and the ladder operators $L_{\pm} = L_x \pm iL_y$ are even-parity operators. Now $Y_l^{m-1} = L_- Y_l^m / \alpha_-$, where α_- is a constant, so applying the parity operator

$$PY_l^{l-1} = \frac{1}{\alpha_-} L_- PY_l^l = (-1)^l \frac{1}{\alpha_-} L_- Y_l^l = (-1)^l Y_l^{l-1}. \quad (7.93)$$

That is, Y_l^{l-1} has the same parity as Y_l^l . Since all the Y_l^m for a given l can be obtained by repeated application of L_- to Y_l^l , it follows that they all have the same parity, $(-1)^l$.

7.4.2 Orbital angular momentum and kinetic energy

We now derive a very useful decomposition of the kinetic-energy operator $H_K = p^2/2m$ into a sum of operators for the radial and tangential kinetic energies. First we show that L^2 is intimately related to the Laplacian operator ∇^2 . From the definition (7.80) of the ladder operators for orbital angular momentum, we have

$$\begin{aligned} L_+ L_- &= (L_x + iL_y)(L_x - iL_y) = L_x^2 + L_y^2 + i[L_y, L_x] \\ &= L_x^2 + L_y^2 + L_z. \end{aligned} \quad (7.94)$$

Hence with equations (7.86) we may write

$$\begin{aligned} L^2 &= L_+ L_- - L_z + L_z^2 \\ &= e^{i\phi} \left(\frac{\partial}{\partial \theta} + i \cot \theta \frac{\partial}{\partial \phi} \right) \left\{ e^{-i\phi} \left(-\frac{\partial}{\partial \theta} + i \cot \theta \frac{\partial}{\partial \phi} \right) \right\} + i \frac{\partial}{\partial \phi} - \frac{\partial^2}{\partial \phi^2}. \end{aligned}$$

Differentiating out the right side

$$L^2 = -\frac{\partial^2}{\partial \theta^2} - \cot^2 \theta \frac{\partial^2}{\partial \phi^2} + \cot \theta \left(-\frac{\partial}{\partial \theta} + i \cot \theta \frac{\partial}{\partial \phi} \right) - i \csc^2 \theta \frac{\partial}{\partial \phi} + i \frac{\partial}{\partial \phi} - \frac{\partial^2}{\partial \phi^2}.$$

The first-order terms in $\partial/\partial\phi$ cancel because $\cot^2 \theta - \csc^2 \theta = -1$. This identity also enables us to combine the double derivatives in ϕ . Finally the

single and double derivatives in θ can be combined so that the equation becomes

$$L^2 = - \left\{ \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right\}, \quad (7.95)$$

which we recognise as $-r^2$ times the angular part of the Laplacian operator ∇^2 .

Now we ask "what is the operator associated with radial momentum?". The obvious candidate is $\hat{\mathbf{r}} \cdot \mathbf{p}$, where $\hat{\mathbf{r}}$ is the unit vector in the radial direction. Unfortunately this operator is not Hermitian:

$$(\hat{\mathbf{r}} \cdot \mathbf{p})^\dagger = \mathbf{p} \cdot \hat{\mathbf{r}} \neq \hat{\mathbf{r}} \cdot \mathbf{p}, \quad (7.96)$$

so it is not an observable. This is a particular case of a general phenomenon: the product AB of two non-commuting observables A and B is never Hermitian. But it is easy to see that $\frac{1}{2}(AB + BA)$ is Hermitian. So we define

$$p_r = \frac{1}{2}(\hat{\mathbf{r}} \cdot \mathbf{p} + \mathbf{p} \cdot \hat{\mathbf{r}}), \quad (7.97)$$

which is manifestly Hermitian. We will need an expression for p_r in the position representation. Replacing $\mathbf{p}$ by $-i\hbar \nabla$ we have

$$p_r = -\frac{i\hbar}{2} \left(\frac{1}{r} \mathbf{r} \cdot \nabla + \nabla \cdot (\mathbf{r}/r) \right). \quad (7.98)$$

From the chain rule it is straightforward to show that

$$\mathbf{r} \cdot \frac{\partial}{\partial \mathbf{r}} = x \frac{\partial}{\partial x} + y \frac{\partial}{\partial y} + z \frac{\partial}{\partial z} = \mathbf{r} \cdot \nabla. \quad (7.99)$$

Moreover, $\nabla \cdot \mathbf{r} = 3$, so equation (7.98) can be rewritten

$$\begin{aligned} p_r &= -\frac{i\hbar}{2} \left(\frac{\partial}{\partial r} + \frac{3}{r} - \frac{r}{r^2} + \frac{\partial}{\partial r} \right) \\ &= -i\hbar \left(\frac{\partial}{\partial r} + \frac{1}{r} \right). \end{aligned} \quad (7.100)$$

This expression enables us to find the commutator

$$[r, p_r] = -i\hbar \left[r, \frac{\partial}{\partial r} \right] = i\hbar. \quad (7.101)$$

Squaring both sides of equation (7.100) yields

$$\begin{aligned} p_r^2 &= -\hbar^2 \left(\frac{\partial}{\partial r} + \frac{1}{r} \right) \left(\frac{\partial}{\partial r} + \frac{1}{r} \right) = -\hbar^2 \left(\frac{\partial^2}{\partial r^2} + \frac{2}{r} \frac{\partial}{\partial r} - \frac{1}{r^2} + \frac{1}{r^2} \right) \\ &= -\frac{\hbar^2}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right). \end{aligned} \quad (7.102)$$

We recognise this operator as $-\hbar^2$ times the radial part of the Laplacian operator ∇^2 . Since we have shown that L^2 is $-r^2$ times the angular part of the Laplacian (eq. 7.95), it follows that $\nabla^2 = -(p_r^2/\hbar^2 + L^2/r^2)$. Consequently, the kinetic-energy operator $H_K = p^2/2m = -(\hbar^2/2m)\nabla^2$ can be written

$$H_K = \frac{1}{2m} \left(p_r^2 + \frac{\hbar^2 L^2}{r^2} \right). \quad (7.103)$$

The physical interpretation of this equation is clear: classically, the orbital angular momentum $\hbar \mathbf{L}$ is $m\mathbf{r} \times \mathbf{v} = mrv_t$, where v_t is the tangential speed, so the term $\hbar^2 L^2/2mr^2 = \frac{1}{2}mv_t^2$ is the kinetic energy associated with the tangential motion. On the other hand $p_r^2/2m = \frac{1}{2}mv_r^2$, so this term represents the kinetic energy due to radial motion, as we would expect. For future reference we note that the kinetic-energy operator can also be written

$$H_K = -\frac{\hbar^2}{2m} \left\{ \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) - \frac{L^2}{r^2} \right\}. \quad (7.104)$$

7.4.3 Legendre polynomials

The spherical harmonic Y_l^0 is special in that it is a function of θ only. We now show that it is, in fact, a polynomial in $\cos\theta$. In the interval $0 \leq \theta \leq \pi$ of interest, θ is a monotone function of $\mu \equiv \cos\theta$, so without any loss of generality we may take Y_l^0 to be proportional to a function $P_l(\mu)$. On this understanding, P_l is an eigenfunction of L^2 with eigenvalue $l(l+1)$. Transforming the independent variable from θ to μ in our expression (7.95) for L^2 , we find that P_l must satisfy **Legendre's equation**:

$$\frac{d}{d\mu} \left((1-\mu^2) \frac{dP_l}{d\mu} \right) + l(l+1)P_l = 0. \quad (7.105)$$

We look for polynomial solutions of this equation. Putting in the trial solution $P_l = \sum_n b_n \mu^n$, we find

$$\sum_n b_n \{ n(n-1)\mu^{n-2} - n(n+1)\mu^n + l(l+1)\mu^n \} = 0. \quad (7.106)$$

This equation must be valid for any value of μ in the interval $(-1, 1)$, which will be possible only if the coefficient of each and every power of μ individually vanishes. The coefficient of μ^k is

$$0 = b_{k+2}(k+2)(k+1) - b_k \{ k(k+1) - l(l+1) \}. \quad (7.107)$$

For $k=0$ the expression connects b_2 to b_0 , while for $k=2$ it relates b_4 to b_2 , and so on. Thus from this equation we can express b_n as a multiple of b_0 for even n , and as a multiple of b_1 for odd n . Moreover, if l is an even number, we know from our discussion of parity that P_l must be an even function of μ , so in this case b_n must vanish for n odd. Finally, b_n will vanish for n even and greater than l on account of the vanishing of the

Table 7.3 The first five Legendre polynomials

	l				
	0	1	2	3	4
$P_l(\mu)$	1	μ	$\frac{1}{2}(3\mu^2 - 1)$	$\frac{1}{2}(5\mu^3 - 3\mu)$	$\frac{1}{8}(35\mu^4 - 30\mu^2 + 3)$

curly bracket in equation (7.107) when $k = l$. This completes the proof that for even l , $P_l(\mu)$ is a polynomial of order l . An extremely similar argument shows that $P_l(\mu)$ is also a polynomial of order l when l is odd. The first five **Legendre polynomials** are listed in Table 7.3.

The conventional normalisation of the Legendre polynomial P_l is the requirement that $P_l(1) = 1$. With this property, the P_l are not orthonormal. In fact

$$\int_{-1}^1 d\mu P_l(\mu)P_{l'}(\mu) = \frac{2}{2l+1}\delta_{ll'}. \quad (7.108)$$

From this result it easily follows that the proportionality constant between $P_l(\cos\theta)$ and the orthonormal functions $Y_l^0(\theta)$ is such that

$$Y_l^0(\theta) = \sqrt{\frac{2l+1}{4\pi}} P_l(\cos\theta). \quad (7.109)$$

7.5 Three-dimensional harmonic oscillator

In this section we discuss the dynamics of a particle that moves in three dimensions subject to a central force that is proportional to the particle's distance from the origin. So the Hamiltonian is

$$H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 r^2. \quad (7.110)$$

If we use Cartesian coordinates, this Hamiltonian becomes the sum of three copies of the Hamiltonian of the one-dimensional harmonic oscillator that was the subject of §3.1:

$$H = H_x + H_y + H_z, \quad (7.111)$$

where, for example, $H_x = (p_x^2/2m) + \frac{1}{2}m\omega^2 x^2$. These one-dimensional Hamiltonians commute with one another. So there is a complete set of mutual eigenkets. Let $|n_x, n_y, n_z\rangle$ be the state that is an eigenket of H_x with eigenvalue $(n_x + \frac{1}{2})\hbar\omega$ (eq. 3.12), etc. Then $|n_x, n_y, n_z\rangle$ will be an eigenket of the three-dimensional Hamiltonian (7.110) with eigenvalue

$$E = (n_x + n_y + n_z + \frac{3}{2})\hbar\omega. \quad (7.112)$$

Moreover, in the position representation the wavefunction of this state is just a product of three of the wavefunctions we derived for stationary states of a one-dimensional oscillator

$$\psi(\mathbf{x}) = u_{n_x}(x)u_{n_y}(y)u_{n_z}(z). \quad (7.113)$$

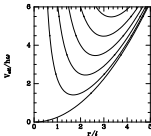


Figure 7.9 The effective potential (7.116) for (from bottom to top) $l = 0, \dots, 5$. The beads mark the classical turning points at the values of energy and angular momentum that quantum mechanics allows.

In view of these considerations it might be thought that there is nothing we do not know about the Hamiltonian (7.110). However, it is instructive to reanalyse the system from a more physical point of view, that recognises that the system is spherically symmetric. We have seen that $[L_z, p^2] = 0$, and $[L_z, r^2] = 0$, so $[L_z, H] = 0$ and $[L^2, H] = 0$. From this result it follows that there is a complete set of mutual eigenstates of H , L^2 and L_z . Very few of the eigenstates obtained from the one-dimensional Hamiltonians are eigenstates of either L^2 or L_z . We now show how the eigenvalue problem associated with (7.110) can be solved in a way that yields mutual eigenkets of H , L^2 and L_z . This exercise is instructive in itself, and some technology that we will develop along the way will prove extremely useful when we analyse the hydrogen atom in Chapter 8.

We use equation (7.103) to eliminate p^2 from equation (7.110)

$$H = \frac{p_r^2}{2m} + \frac{\hbar^2 L^2}{2mr^2} + \frac{1}{2}m\omega^2 r^2. \quad (7.114)$$

We can assume that our energy eigenstates are also eigenstates of L^2 , so in this Hamiltonian we can replace L^2 by an eigenvalue $l(l+1)$. Hence we wish to find the eigenvalues of the **radial Hamiltonian**

$$H_l = \frac{p_r^2}{2m} + \frac{l(l+1)\hbar^2}{2mr^2} + \frac{1}{2}m\omega^2 r^2. \quad (7.115)$$

This is the Hamiltonian for a particle that moves in a one-dimensional **effective potential**

$$V_{\text{eff}}(r) = \frac{l(l+1)\hbar^2}{2mr^2} + \frac{1}{2}m\omega^2 r^2. \quad (7.116)$$

H_l governs the oscillations of the mass about the minima of this potential, which is plotted in Figure 7.9. The eigenkets $|E, l, m\rangle$ of H are products of the eigenkets $|E, l\rangle$ of H_l and the eigenkets $|l, m\rangle$ of L^2 and L_z :

$$|E, l, m\rangle = |E, l\rangle |l, m\rangle. \quad (7.117)$$

Our determination of the allowed energies of a one-dimensional harmonic oscillator exploited the dimensionless operators A and $A^\dagger$, which rather nearly factorise $H/\hbar\omega$. So here we define the operator

$$A_l = \frac{1}{\sqrt{2m\hbar\omega}} \left(ip_r - \frac{(l+1)\hbar}{r} + m\omega r \right). \quad (7.118)$$

The product of A and its Hermitian adjoint $A_l^\dagger$ is

$$\begin{aligned} A_l^\dagger A_l &= \frac{1}{2m\hbar\omega} \left(-ip_r - \frac{(l+1)\hbar}{r} + m\omega r \right) \left(ip_r - \frac{(l+1)\hbar}{r} + m\omega r \right) \\ &= \frac{1}{2m\hbar\omega} \left\{ p_r^2 + \left(-\frac{(l+1)\hbar}{r} + m\omega r \right)^2 + i \left[-\frac{(l+1)\hbar}{r} + m\omega r, p_r \right] \right\} \\ &= \frac{1}{2m\hbar\omega} \left\{ p_r^2 + \frac{l(l+1)\hbar^2}{r^2} + m^2\omega^2 r^2 - (2l+3)\hbar m\omega \right\}. \end{aligned} \quad (7.119)$$

Comparing the right side with equation (7.115), we see that

$$H_l = \hbar\omega \left\{ A_l^\dagger A_l + \left(l + \frac{3}{2} \right) \right\}, \quad (7.120)$$

which bears a strong similarity to equation (3.3) for the one-dimensional harmonic oscillator.

The commutator of A_l and $A_l^\dagger$ is

$$\begin{aligned} [A_l, A_l^\dagger] &= \frac{1}{2m\hbar\omega} \left[\left(ip_r - \frac{(l+1)\hbar}{r} + m\omega r \right), \left(-ip_r - \frac{(l+1)\hbar}{r} + m\omega r \right) \right] \\ &= \frac{i}{m\hbar\omega} \left[p_r, \left(-\frac{(l+1)\hbar}{r} + m\omega r \right) \right] \\ &= \frac{(l+1)\hbar}{m\omega r^2} + 1, \end{aligned} \quad (7.121)$$

where we have used equation (7.101) to reach the last line. This result can be written more usefully in the form

$$[A_l, A_l^\dagger] = \frac{H_{l+1} - H_l}{\hbar\omega} + 1. \quad (7.122)$$

From this expression and equation (7.120) we can easily calculate the commutator of H_l with A_l :

$$[A_l, H_l] = \hbar\omega [A_l, A_l^\dagger A_l] = \hbar\omega [A_l, A_l^\dagger] A_l = (H_{l+1} - H_l + \hbar\omega) A_l. \quad (7.123)$$

Now let $|E, l\rangle$ be an eigenket of H_l with eigenvalue E :

$$H_l |E, l\rangle = E |E, l\rangle. \quad (7.124)$$

We multiply both sides of the equation by A_l and use equation (7.123) to reverse the order of A_l and H_l :

$$\begin{aligned} EA_l|E, l\rangle &= A_l H_l|E, l\rangle = (H_l A_l + [A_l, H_l])|E, l\rangle \\ &= (H_{l+1} + \hbar\omega)A_l|E, l\rangle. \end{aligned} \quad (7.125)$$

On rearrangement this yields

$$H_{l+1}(A_l|E, l\rangle) = (E - \hbar\omega)(A_l|E, l\rangle), \quad (7.126)$$

which says that $A_l|E, l\rangle$ is an eigenket of H_{l+1} for the eigenvalue $E - \hbar\omega$, so

$$A_l|E, l\rangle = \alpha_-|E - \hbar\omega, l+1\rangle, \quad (7.127)$$

where α_- is a normalising constant.

A_l creates the radial wavefunction for a state that has *more* orbital angular momentum and *less* energy than the state with which it started. That is, A_l diminishes the radial kinetic energy by some amount and adds a smaller quantity of energy to the tangential motion. If we repeat this process a sufficient number of times, by following A_l with A_{l+1} and A_{l+1} with A_{l+2} , and so on, there will come a point at which no radial kinetic energy remains – we will have reached the quantum equivalent of a circular orbit. The next application of A_l must annihilate the wavefunction. Hence $A_{\mathcal{L}}|E, \mathcal{L}\rangle = 0$, where $\mathcal{L}(E)$ is the largest allowed value of l for energy E . If we operate on $|E, \mathcal{L}\rangle$ with $H_{\mathcal{L}}$, we find with equation (7.120) that

$$E|E, \mathcal{L}\rangle = H_{\mathcal{L}}|E, \mathcal{L}\rangle = \hbar\omega(\mathcal{L} + \tfrac{3}{2})|E, \mathcal{L}\rangle, \quad (7.128)$$

so

$$E = (\mathcal{L} + \tfrac{3}{2})\hbar\omega \quad \text{and} \quad \mathcal{L}(E) = \frac{E}{\hbar\omega} - \tfrac{3}{2}.$$

Since $\mathcal{L}$ is a non-negative integer, it follows that the ground-state energy is $\frac{3}{2}\hbar\omega$ and that the ground state has no angular momentum. In general $E/\hbar\omega$ is any integer plus $\frac{3}{2}$. These values of the allowed energies agree perfectly with what we could have deduced by treating H as a sum of three one-dimensional harmonic-oscillator Hamiltonians.

We shall define a **circular orbit** to be one that has the maximum angular momentum possible at its energy. We obtain the radial wavefunctions of these by writing the equation $A_{\mathcal{L}}|E, \mathcal{L}\rangle = 0$ in the position representation. With equations (7.118) and (7.100) this equation becomes

$$\left(\frac{\partial}{\partial r} + \frac{1}{r} - \frac{\mathcal{L}+1}{r} + \frac{m\omega}{\hbar} r \right) u_{\mathcal{L}}(r) = 0. \quad (7.129)$$

This is a first-order linear differential equation. Its integrating factor is

$$\exp \left\{ \int dr \left(-\frac{\mathcal{L}}{r} + \frac{m\omega}{\hbar} r \right) \right\} = r^{-\mathcal{L}} \exp \left(\frac{m\omega}{2\hbar} r^2 \right), \quad (7.130)$$

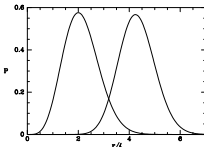


Figure 7.10 Radial probability distributions of circular orbits in the three-dimensional harmonic oscillator potential for $\mathcal{L} = 1$ and $\mathcal{L} = 8$. The scale radius $\ell = \sqrt{\hbar/2m\omega}$.

so the solution of equation (7.129) is

$$u_{\mathcal{L}}(r) = \text{constant} \times r^{\mathcal{L}} e^{-r^2/4\ell^2}, \quad \text{where} \quad \ell = \sqrt{\hbar/2m\omega}. \quad (7.131)$$

Notice that the exponential factor is simply the product of three exponential factors from equation (3.15), one in x , one in y and one in z . The wavefunction varies with r , so a circular orbit does have some radial kinetic energy. In the limit of large $\mathcal{L}$ in which classical mechanics applies, the radial kinetic energy is negligible compared to the tangential kinetic energy, and we neglect it. But it never really vanishes.

Equation (7.131) gives the radial wavefunction for a circular orbit. The complete wavefunction is $\psi(\mathbf{x}) = u_{\mathcal{L}}(r)Y_{\mathcal{L}}^m(\theta, \phi)$, and since the spherical harmonics satisfy $\int d^2\Omega |Y_{\mathcal{L}}^m|^2 = 1$, the radial probability density is $P(r) = r^2 u_{\mathcal{L}}^2 \propto r^{2\mathcal{L}+2} e^{-r^2/2\ell^2}$, where the factor r^2 arises from the expression for the volume element $d^3\mathbf{x}$ in spherical polar coordinates. This density is plotted in Figure 7.10 for $\mathcal{L} = 1$ and $\mathcal{L} = 8$. For $r/\ell \ll \sqrt{2\mathcal{L}+2}$, P rises as $r^{2\mathcal{L}+2}$. For $r/\ell > \sqrt{2\mathcal{L}+2}$ it falls rapidly as the Gaussian factor takes over. Figure 7.10 shows that the uncertainty in r is $\sim \ell$, which is a small fraction of r when $\mathcal{L}$ is not small.

We may obtain the radial wavefunctions of more eccentric orbits by showing that $A_l^\dagger$ is a raising operator. Equation (7.123) yields

$$A_l H_l = H_{l+1} A_l + \hbar\omega A_l. \quad (7.132)$$

Daggering both sides we have

$$H_l A_l^\dagger = A_l^\dagger H_{l+1} + \hbar\omega A_l^\dagger. \quad (7.133)$$

We now multiply both sides of $H_{l+1}|E, l+1\rangle = E|E, l+1\rangle$ by $A_l^\dagger$:

$$E A_l^\dagger |E, l+1\rangle = A_l^\dagger H_{l+1} |E, l+1\rangle = (H_l - \hbar\omega) A_l^\dagger |E, l+1\rangle. \quad (7.134)$$

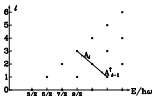


Figure 7.11 The (E, l) plane and the action of A_l and $A_{l-1}^\dagger$.

Rearranging

$$H_l(A_l^\dagger|E, l+1) = (E + \hbar\omega)(A_l^\dagger|E, l+1). \quad (7.135)$$

Thus, we have shown that

$$A_l^\dagger|E, l+1 = \alpha_+|E + \hbar\omega, l\rangle, \quad (7.136)$$

where α_+ is a normalising constant. By writing $A_l^\dagger$ in the position representation, we can generate the wavefunctions of all non-circular orbits by repeatedly applying $A_l^\dagger$ to the current wavefunction, starting with that of a circular orbit. We start with the product of r^ℓ and a Gaussian factor (eq. 7.131). From this $A_{\ell-1}^\dagger$ generates terms proportional to $r^{\ell+1}$ and $r^{\ell-1}$ times the Gaussian (Problem 7.25). From these two terms $A_{\ell-2}^\dagger$ then generates three terms, $r^{\ell+2}$, r^ℓ and $r^{\ell-2}$ times the Gaussian, and so on. Consequently the number of radial **nodes** – radii at which the wavefunction vanishes – increases by one with each application of $A_l^\dagger$, and the wavefunction oscillates more and more rapidly in radius as $A_l^\dagger$ invests a larger and larger fraction of the particle's kinetic energy in radial motion.

Figure 7.11 helps to organise the generation of radial wavefunctions. Each dot represents a radial wavefunction. From the dot at (E, l) , operating with A_l carries one to the next dot up and to the left, while operating with $A_{l-1}^\dagger$ carries one to the next dot down and to the right. At half the energies only even values of l occur, and only odd values of l occur at the other half of the energies. In Problem 7.22 you can show that, when one bears in mind that each dot gives rise to $2l + 1$ complete wavefunctions, the number of wavefunction with energy E that we obtain in this way agrees with the number that we would obtain using wavefunctions of the one-dimensional harmonic oscillator via equation (7.113).

7.6 Addition of angular momenta

In practical applications of quantum mechanics we can often identify two or more components of the system that each carry a well-defined amount of angular momentum. For example, in a hydrogen atom both the proton and the electron carry angular momentum $\frac{1}{2}\hbar$ by virtue of their spins, and a further quantity of angular momentum may be present in the orbit of the electron around the proton. The total angular momentum of the atom is the sum of these three contributions, so it is important to understand how to add angular momenta in quantum mechanics. Once we understand how to add two contributions, we'll be able to add any number of contributions, because we can add the third contribution to the result of adding the first two, and so on. Therefore in this section we focus on the problem of adding the angular momenta of two 'gyros', that is two systems that have unvarying total angular-momentum quantum number j but several possible orientations.

Imagine that we have two gyros in a box and that we know that the first gyro has total angular-momentum quantum number j_1 , while the second gyro has total quantum number j_2 . Without loss of generality we may assume $j_1 \geq j_2$. A ket describing the state of the first gyro is of the form

$$|\psi_1\rangle = \sum_{m=-j_1}^{j_1} c_m |j_1, m\rangle, \quad (7.137a)$$

while the state of the second is

$$|\psi_2\rangle = \sum_{m=-j_2}^{j_2} d_m |j_2, m\rangle, \quad (7.137b)$$

and from the discussion in §6.1 it follows that the state of the box is

$$|\psi\rangle = |\psi_1\rangle |\psi_2\rangle. \quad (7.138)$$

The coefficients c_m and d_m are the amplitudes to find the individual gyros in particular orientations with respect to the z -axis. For example, if both gyros are maximally aligned with the z -axis, we will have $|c_{j_1}| = |d_{j_2}| = 1$ and $c_{m_1} = d_{m_2} = 0$ for $m_1 \neq j_1$ and $m_2 \neq j_2$.

The operators of interest are the operators J_i^2 , J_{iz} and $J_{i\pm}$ of the i^{th} gyro and the corresponding operators of the box. The operators J_z and $J_{\pm}$ for the box are simply sums of the corresponding operators for the gyros

$$J_z = J_{1z} + J_{2z}, \quad J_{\pm} = J_{1\pm} + J_{2\pm}. \quad (7.139)$$

Operators belonging to different systems always commute, so $[J_{1i}, J_{2j}] = 0$ for any values of i, j . The operator for the square of the box's angular momentum is

$$J^2 = (\mathbf{J}_1 + \mathbf{J}_2)^2 = J_1^2 + J_2^2 + 2\mathbf{J}_1 \cdot \mathbf{J}_2. \quad (7.140)$$

Now

$$\begin{aligned} J_{1+}J_{2-} &= (J_{1x} + iJ_{1y})(J_{2x} - iJ_{2y}) \\ &= (J_{1x}J_{2x} + J_{1y}J_{2y}) + i(J_{1y}J_{2x} - J_{1x}J_{2y}). \end{aligned} \quad (7.141)$$

The expression for $J_{1-}J_{2+}$ can be obtained by swapping the labels 1 and 2, so⁹

$$J_{1+}J_{2-} + J_{1-}J_{2+} + 2J_{1z}J_{2z} = 2\mathbf{J}_1 \cdot \mathbf{J}_2. \quad (7.142)$$

Using this expression to eliminate $\mathbf{J}_1 \cdot \mathbf{J}_2$ from (7.140) we obtain

$$J^2 = J_1^2 + J_2^2 + J_{1+}J_{2-} + J_{1-}J_{2+} + 2J_{1z}J_{2z}. \quad (7.143)$$

While the total angular momenta of the individual gyros are fixed, that of the box is variable because it depends on the mutual orientation of the two gyros: if the latter are parallel, the squared angular momentum in the box might be expected to have quantum number $j_1 + j_2$, while if they are antiparallel, the box's angular momentum might be expected to have quantum number $j_1 - j_2$. We shall show that this conjecture is true by explicitly calculating the values of the coefficients c_m and d_m for which the box is in an eigenstate of both J^2 and J_z . We start by examining the state $|j_1, j_1\rangle|j_2, j_2\rangle$ in which both gyros are maximally aligned with the z -axis. It is easy to see that this object is an eigenket of J_z with eigenvalue $j_1 + j_2$. We use (7.143) to show that it is also an eigenket of J^2 :

$$\begin{aligned} J^2|j_1, j_1\rangle|j_2, j_2\rangle &= (J_1^2 + J_2^2 + J_{1+}J_{2-} + J_{1-}J_{2+} + 2J_{1z}J_{2z})|j_1, j_1\rangle|j_2, j_2\rangle \\ &= \{j_1(j_1 + 1) + j_2(j_2 + 1) + 2j_1j_2\}|j_1, j_1\rangle|j_2, j_2\rangle, \end{aligned} \quad (7.144)$$

where we have used the equation $J_{\pm}|j_i, j_i\rangle = 0$, which follows from equation (7.7). Now

$$\begin{aligned} j_1(j_1 + 1) + j_2(j_2 + 1) + 2j_1j_2 &= j_1(j_1 + j_2 + 1) + j_2(j_1 + j_2 + 1) \\ &= (j_1 + j_2)(j_1 + j_2 + 1) \end{aligned} \quad (7.145)$$

so the expression in curly brackets in equation (7.144) equals $j(j+1)$ with $j = j_1 + j_2$. Hence $|j_1, j_1\rangle|j_2, j_2\rangle$ satisfies both the defining equations of the state $|j_1 + j_2, j_1 + j_2\rangle$ and we may write

$$|j_1 + j_2, j_1 + j_2\rangle = |j_1, j_1\rangle|j_2, j_2\rangle. \quad (7.146)$$

Now that we have found one mutual eigenket for the box of J^2 and J_z we can easily find others by applying $J_{\pm}$ to reorient the angular momentum of the box away from the z -axis. Again setting $j = j_1 + j_2$ we evaluate the two sides of the equation

$$J_{\pm}|j, j\rangle = (J_{1\pm} + J_{2\pm})|j_1, j_1\rangle|j_2, j_2\rangle. \quad (7.147)$$

⁹ Recall that J_{1i} commutes with J_{2j} for all ij .

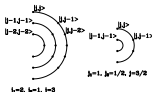


Figure 7.12 The left panel shows states obtained by adding a system of angular momentum $j_2 = 1$ to one with $j_1 = 2$, while the right panel is for $j_1 = 1$ and $j_2 = \frac{1}{2}$.

Equation (7.7) enables us to rewrite the left side

$$J_- |j, j\rangle = \sqrt{j(j+1) - j(j-1)} |j, j-1\rangle = \sqrt{2j} |j, j-1\rangle. \quad (7.148)$$

The right side of (7.147) becomes

$$\begin{aligned} & \sqrt{j_1(j_1+1) - j_1(j_1-1)} |j_1, j_1-1\rangle |j_2, j_2\rangle \\ & + \sqrt{j_2(j_2+1) - j_2(j_2-1)} |j_1, j_1\rangle |j_2, j_2-1\rangle \\ & = \sqrt{2j_1} |j_1, j_1-1\rangle |j_2, j_2\rangle + \sqrt{2j_2} |j_1, j_1\rangle |j_2, j_2-1\rangle. \end{aligned} \quad (7.149)$$

Putting the two sides back together, we have

$$|j, j-1\rangle = \sqrt{\frac{j_1}{j}} |j_1, j_1-1\rangle |j_2, j_2\rangle + \sqrt{\frac{j_2}{j}} |j_1, j_1\rangle |j_2, j_2-1\rangle. \quad (7.150)$$

A further application of J_- to the left side of this equation and of $J_{1-} + J_{2-}$ to the right side would produce an expression for $|j, j-2\rangle$ and so on.

Figure 7.12 helps to organise the results of this calculation. States of the box with well-defined angular momentum are marked by dots. The radius of each semicircle is proportional to j' , where $j'(j'+1)$ is the eigenvalue of the kets with respect to J^2 . The height of each ket above the centre of the circles is proportional to m . The left panel shows the case $j_1 = 2$, $j_2 = 1$, while the right panel is for $j_1 = 1$, $j_2 = \frac{1}{2}$. The scheme for constructing eigenstates J^2 and J_z that we have developed so far starts with the state at the top and then uses J_- to successively generate the states that lie on the outermost semicircle.

We now seek an expression for the state $|j-1, j-1\rangle$ that lies at the top of the first semicircle inwards. It is trivial to verify that $|j_1, m_1\rangle |j_2, m_2\rangle$ is an eigenket of J_z with eigenvalue $(m_1 + m_2)$. We require $m_1 + m_2 = j_1 + j_2 - 1$, so either $m_1 = j_1 - 1$ and $m_2 = j_2$, or $m_1 = j_1$ and $m_2 = j_2 - 1$. Equation (7.150) shows that $|j, j-1\rangle$ involves precisely these two cases, and must be orthogonal to $|j-1, j-1\rangle$ because it has a different eigenvalue with respect to J^2 . So the ket we seek is the unique (up to an overall

Table 7.4 Total numbers of states

j	Number of states
$j_1 + j_2$	$2(j_1 + j_2) + 1$
$j_1 + j_2 - 1$	$2(j_1 + j_2) + 1 - 2$
...	...
$j_1 - j_2$	$2(j_1 + j_2) + 1 - 4j_2$
Total	$(2j_1 + 1)(2j_2 + 1)$

phase factor) linear combination of the kets appearing in (7.150) that is orthogonal to the linear combination that appears there. That is,

$$|j-1, j-1\rangle = \sqrt{\frac{j_2}{j}} |j_1, j_1-1\rangle |j_2, j_2\rangle - \sqrt{\frac{j_1}{j}} |j_1, j_1\rangle |j_2, j_2-1\rangle. \quad (7.151)$$

All the kets $|j-1, m\rangle$ for $m = j-2, \dots$, which in Figure 7.12 lie on the first semicircle in, can be constructed by applying J_- to this equation.

Similarly, $|j-2, j-2\rangle$, which in Figure 7.12 lies at the top of the smallest semicircle, will be a linear combination of $|j_1, j_1-2\rangle |j_2, j_2\rangle$, $|j_1, j_1-1\rangle |j_2, j_2-1\rangle$ and $|j_1, j_1\rangle |j_2, j_2-2\rangle$ and must be orthogonal to $|j, j-2\rangle$ and $|j-1, j-2\rangle$, which are known linear combinations of these states. Hence we can determine which linear combination is required for $|j-2, j-2\rangle$, and then generate the remaining kets of the series $|j-2, m\rangle$ by applying J_- to it.

On physical grounds we would expect the box's smallest total angular-momentum quantum number to be $j_1 - j_2$, corresponding to the case in which the two gyros are antiparallel (recall that we have labelled the gyros such that $j_1 \geq j_2$). Does this conjectured smallest value of j allow for the correct number of basis states for the box? That is, will there be as many basis states of the box as there are of the contents of the box? We can easily evaluate the latter: there are $2j_1 + 1$ orientations of the first gyro, and for each of these orientations, the second gyro can be oriented in $2j_2 + 1$ ways. So the box's contents can be in $(2j_1 + 1)(2j_2 + 1)$ basis states of the form $|j_1, m_1\rangle |j_2, m_2\rangle$. The predicted number of basis states of the box is worked out in Table 7.4. In the main part of the table, the number of states in each row is two less than in the row above and there are $2j_2 + 1$ rows. The sum at the bottom can be obtained by making a third column that is just the second column in reverse order and noting that the sum of the numbers in the second and third columns of a given row is then always $4j_1 + 2$. Hence twice the sum of the numbers in the second column is $2j_2 + 1$ times $4j_1 + 2$. Thus we *do* get the correct number of basis states if the smallest value of j is $j_1 - j_2$.

The numbers

$$C(j, m; j_1, j_2, m_1, m_2) \equiv \langle j, m | j_1, m_1 \rangle |j_2, m_2\rangle \quad (7.152)$$

that we have been evaluating are called **Clebsch-Gordan** coefficients. They have a simple physical interpretation: $C(j, m; j_1, j_2, m_1, m_2)$ is the

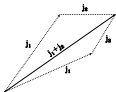


Figure 7.13 Interpretation of Clebsch-Gordan coefficients in terms of vectors. The full line has length $\sqrt{3(3+1)}$ and its vertical component has length 2. The dotted lines labelled j_1 have length $\sqrt{2(2+1)}$ and vertical components of length 2 and 1.

amplitude that, on opening the box when it's in a state of well-defined angular momentum, we will find the first and second gyros to be oriented with amounts m_1 and m_2 of their spins parallel to the z -axis. For example, equation (7.150) implies that $C(3, 2; 2, 1, 1, 1) = \sqrt{2/3}$, so if a box that contains a spin-two gyro and a spin-one gyro has spin-three, there is a probability $2/3$ that on opening the box the second gyro will be maximally aligned with the z -axis and the second significantly inclined, and only a probability $1/3$ of finding the reverse arrangement. These two possibilities are depicted by the lower and upper dotted lines in Figure 7.13. The classical interpretation is that the two gyros precess around the fixed angular-momentum vector of the box, and that the two configurations for which the Clebsch-Gordan coefficients give amplitudes are two of the states through which the precession carries the system. This picture is intuitive and of some value, but should not be taken too seriously. For one thing, the rules for adding angular momentum are independent of any statement about the Hamiltonian, and therefore carry no implication about the time evolution of the system. The gyros may or may not precess, depending on whether they are dynamically coupled.

In §6.1 we saw that the physical significance of the state of a composite system, such as that formed by two gyros, being a linear combination of product states such as $|j_1, m_1\rangle|j_2, m_2\rangle$, is that the subsystems are correlated. The Clebsch-Gordan coefficients encode the correlations between the gyros required for the box to have well-defined angular momentum. If there is any uncertainty in the orientation of either gyro, such correlations are essential if the angular momentum of the box is to be well-defined: the angular momentum of the second gyro has to make up a pre-defined total with whatever value is measured for the first gyro. This consideration explains why the only states of the box that are simple products of states of the individual gyros are $|j_1 + j_2, j_1 + j_2\rangle = |j_1, j_1\rangle|j_2, j_2\rangle$ and $|j_1 + j_2, -(j_1 + j_2)\rangle = |j_1, -j_1\rangle|j_2, -j_2\rangle$ – so much angular momentum can be aligned with the z -axis only by each gyro individually straining to align with the axis, and there is then no need for the gyros to coordinate their efforts.

7.6.1 Case of two spin-half systems

The general analysis we have just given will be clarified by working out some particular cases. We consider first the case $j_1 = j_2 = \frac{1}{2}$, which

is relevant, for example, to a hydrogen atom in its ground state, when all angular momentum is contributed by the spins of the proton and the electron. The electron has base states $|\pm, e\rangle$ in which J_z returns the value $\pm\frac{1}{2}$, while the proton has corresponding base states $|\pm, p\rangle$. Hence there are four states in all and j takes just two values, 1 and 0.

Our construction of the states in which the atom has well-defined angular momentum starts with the state

$$|1, 1\rangle = |+, e\rangle|+, p\rangle \quad (7.153)$$

in which both the electron and the proton have their spins maximally aligned with the z -axis. So the atom has maximum angular momentum, and its angular momentum is maximally aligned with the z -axis. Applying $J_- = J_-^e + J_-^p$ to this ket we obtain

$$|1, 0\rangle = \frac{1}{\sqrt{2}} (|-, e\rangle|+, p\rangle + |+, e\rangle|-, p\rangle). \quad (7.154)$$

The right side of this equation states that with the atom in this state, measurements of J_z for the electron and proton are certain to find that they are 'antiparallel'. This fact is surprising given that the left side states that the atom has maximum angular momentum, so you would think that the two particles had parallel angular momenta. The resolution of this paradox is that the z components of the two spins are antiparallel, but the components in the xy plane are parallel, although their direction is unknown to us. Similarly, when the atom is in the state $|1, 1\rangle$ of equation (7.153), the z components of the electron and proton angular momenta are parallel, but the components in the xy plane are not well aligned. The poor alignment in the xy plane explains why $\sqrt{J^2} = \sqrt{2}$ for the atom is less than $\sqrt{3}$, which is the sum of $\sqrt{J^2} = \sqrt{3/4}$ for the electron and the proton.

When we apply J_- to $|1, 0\rangle$ we obtain

$$|1, -1\rangle = |-, e\rangle|-, p\rangle. \quad (7.155)$$

This equation confirms the physically obvious fact that if we want to have $\hbar$ of angular momentum pointing along the negative z -axis, we need to have the angular momenta of both the proton and the electron maximally aligned with the negative z -axis.

The remaining state of the atom is $|0, 0\rangle$ in which the atom has no angular momentum. This is the unique linear combination of the two compound states on the right of equation (7.154) that is orthogonal to $|1, 0\rangle$:

$$|0, 0\rangle = \frac{1}{\sqrt{2}} (|-, e\rangle|+, p\rangle - |+, e\rangle|-, p\rangle). \quad (7.156)$$

The change of sign on the right of this equation from the right of equation (7.154) for $|1, 0\rangle$ ensures that the spins of the electron and proton are antiparallel in the xy plane as well as along the z -axis. We show this by

rewriting $|1, 0\rangle$ and $|0, 0\rangle$ in terms of the states in which the electron and proton have well-defined spin parallel to the x -axis. These states are

$$\begin{aligned} |x+, e\rangle &= \frac{1}{\sqrt{2}} (|+, e\rangle + |-, e\rangle) & |x-, e\rangle &= \frac{1}{\sqrt{2}} (|+, e\rangle - |-, e\rangle) \\ |x+, p\rangle &= \frac{1}{\sqrt{2}} (|+, p\rangle + |-, p\rangle) & |x-, p\rangle &= \frac{1}{\sqrt{2}} (|+, p\rangle - |-, p\rangle) \end{aligned} \quad (7.157)$$

So

$$\begin{aligned} |0, 0\rangle &= |x+, e\rangle \langle x+, e|0, 0\rangle + |x-, e\rangle \langle x-, e|0, 0\rangle \\ &= \frac{1}{2} |x+, e\rangle (-|-, p\rangle + |+, p\rangle) - \frac{1}{2} |x-, e\rangle (|-, p\rangle + |+, p\rangle) \\ &= -\frac{1}{\sqrt{2}} (|x+, e\rangle |x-, p\rangle + |x-, e\rangle |x+, p\rangle). \end{aligned} \quad (7.158)$$

The last line states that when the atom is in the state $|0, 0\rangle$ we are indeed guaranteed to find the components of the spins of the electron and proton parallel to x have opposite signs. An analogous calculation starting from equation (7.154) yields (Problem 7.27)

$$|1, 0\rangle = \frac{1}{\sqrt{2}} (|x+, e\rangle |x+, p\rangle - |x-, e\rangle |x-, p\rangle), \quad (7.159)$$

so when the atom is in the $|1, 0\rangle$ state the two particles have identical components of spin along x .

Notice that all three states in which the atom has $j = 1$ are unchanged if we swap the m values of the particles – that is, if we map $|\pm, e\rangle \rightarrow |\mp, e\rangle$ and the same for the proton states. The atomic state with $j = 0$, by contrast, changes sign under this interchange. This fact will prove to be important when we consider systems with two electrons (such as helium) or two protons (such as an H_2 molecule).

7.6.2 Case of spin-one and spin-half

In the first excited state of hydrogen, the electron can have total orbital angular-momentum quantum number $l = 1$. So we now consider how to combine angular momentum $j = 1$ with the electron's spin, $j = \frac{1}{2}$. The total angular-momentum quantum number takes two values, $j = \frac{3}{2}$ and $j = \frac{1}{2}$ (see Figure 7.12). We start with the state

$$|\frac{3}{2}, \frac{3}{2}\rangle = |+\rangle|1, 1\rangle \quad (7.160)$$

in which the spin and orbital angular momenta are both maximally oriented along the z -axis. Applying $J_- = L_- + S_-$ to this equation, we obtain

$$|\frac{3}{2}, \frac{1}{2}\rangle = \sqrt{\frac{2}{3}} |-\rangle|1, 1\rangle + \sqrt{\frac{1}{3}} |+\rangle|1, 0\rangle. \quad (7.161)$$

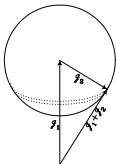


Figure 7.14 Classically the sum vector $\mathcal{J}_1 + \mathcal{J}_2$ can lie anywhere on the sphere of radius $|\mathcal{J}_2|$ around the end of $\mathcal{J}_1$.

The right side of this equation says that in this state of the atom, the electron is twice as likely to be found with its spin up as down. A second application of J_- yields

$$|\tfrac{3}{2}, -\tfrac{1}{2}\rangle = \sqrt{\tfrac{2}{3}}|-\rangle|1, 0\rangle + \sqrt{\tfrac{1}{3}}|+\rangle|1, -1\rangle \quad (7.162)$$

as we would expect from the symmetry between up and down. A final application of J_- yields $|\tfrac{3}{2}, -\tfrac{3}{2}\rangle = |-\rangle|1, -1\rangle$ as it must on physical grounds.

The state $|\tfrac{1}{2}, \tfrac{1}{2}\rangle$ is the linear combination of the states that appear in the right of equation (7.161) that is orthogonal to $|\tfrac{3}{2}, \tfrac{1}{2}\rangle$. Hence,

$$|\tfrac{1}{2}, \tfrac{1}{2}\rangle = \sqrt{\tfrac{2}{3}}|-\rangle|1, 1\rangle - \sqrt{\tfrac{1}{3}}|+\rangle|1, 0\rangle. \quad (7.163)$$

In this atomic state, the electron's spin is twice as likely to be down as up. The last remaining state can be found by applying J_- to equation (7.163). It is

$$|\tfrac{1}{2}, -\tfrac{1}{2}\rangle = \sqrt{\tfrac{1}{3}}|-\rangle|1, 0\rangle - \sqrt{\tfrac{2}{3}}|+\rangle|1, -1\rangle. \quad (7.164)$$

7.6.3 The classical limit

In classical physics we identify angular momentum with the vector $\mathcal{J} = \hbar \langle \mathbf{J} \rangle$, and the angular momentum of the whole system is obtained by vectorially adding the angular momenta $\mathcal{J}_1$ and $\mathcal{J}_2$ of the component parts. If θ is the angle between these vectors, then

$$\mathcal{J}^2 = \mathcal{J}_1^2 + \mathcal{J}_2^2 + 2\mathcal{J}_1\mathcal{J}_2 \cos \theta. \quad (7.165)$$

If nothing is known about the direction of $\mathcal{J}_2$ relative to $\mathcal{J}_1$, all points on a sphere of radius $\mathcal{J}_2$ and centred on the end of $\mathcal{J}_1$ are equally likely locations for the end of $\mathcal{J}_2$ (Figure 7.14). Consequently, the probability dP that θ lies in the range $(\theta, \theta + d\theta)$ is proportional to the area of the band shown in the figure. Quantitatively

$$dP = \frac{1}{2} \sin \theta d\theta, \quad (7.166)$$

where the factor $\frac{1}{2}$ ensures that $\int dP = 1$. From equation (7.165) the change in $\mathcal{J}$ when θ changes by $d\theta$ is given by

$$\mathcal{J} d\mathcal{J} = -\mathcal{J}_1 \mathcal{J}_2 \sin \theta d\theta. \quad (7.167)$$

Combining equations (7.166) and (7.167), we find that the probability that the total angular momentum lies in the interval $(\mathcal{J}, \mathcal{J} + d\mathcal{J})$ is¹⁰

$$dP = \frac{\mathcal{J} d\mathcal{J}}{2\mathcal{J}_1 \mathcal{J}_2}. \quad (7.168)$$

In quantum mechanics the fraction of states that have total angular-momentum quantum number j is

$$f = \frac{2j+1}{(2j_1+1)(2j_2+1)}, \quad (7.169)$$

which in the classical limit of large quantum numbers becomes approximately $j/(2j_1 j_2)$. If all states were equally likely, this fraction would equal the classical probability that $\mathcal{J} \simeq j\hbar$ lay within $\hbar$ of $j\hbar$. It is easy to check from (7.167) that dP does indeed take the value f when we insert $\mathcal{J}_i = \hbar j_i$ and $d\mathcal{J} = \hbar$. Thus from consistency with classical mechanics we are led to the **principle of equal a priori probability**, namely that *when we have no information relevant to an upcoming measurement, we assign equal probabilities to the system being in each state of whatever basis we have decided to work in*. This principle is the foundation of all statistical physics.

Problems

7.1 Show that $\langle j, j | J_x | j, j \rangle = \langle j, j | J_y | j, j \rangle = 0$ and that $\langle j, j | (J_x^2 + J_y^2) | j, j \rangle = j$. Discuss the implications of these results for the uncertainty in the orientation of the classical angular momentum vector $\mathbf{J}$ for both small and large values of j .

7.2 In the rotation spectrum of $^{12}\text{C}^{16}\text{O}$ the line arising from the transition $l = 4 \rightarrow 3$ is at 461.04077 GHz, while that arising from $l = 36 \rightarrow 35$ is at 4115.6055 GHz. Show from these data that in a non-rotating CO molecule the intra-nuclear distance is $s \simeq 0.113 \text{ nm}$, and that the electrons provide a spring between the nuclei that has force constant $\sim 1904 \text{ N m}^{-1}$. Hence show that the vibrational frequency of CO should lie near $6.47 \times 10^{13} \text{ Hz}$ (measured value is $6.43 \times 10^{13} \text{ Hz}$). Hint: show from classical mechanics that the distance of O from the centre of mass is $\frac{3}{7}s$ and that the molecule's moment of inertia is $\frac{48}{7}m_p s^2$. Recall also the classical relation $L = I\omega$.

¹⁰ We discarded the minus sign in equation (7.167) because we require $dP > 0$ regardless of whether $\mathcal{J}$ increases or decreases.

7.3 Show that L_i commutes with $\mathbf{x} \cdot \mathbf{p}$ and thus also with scalar functions of $\mathbf{x}$ and $\mathbf{p}$.

7.4 Write down the expression for the commutator $[\sigma_i, \sigma_j]$ of two Pauli matrices. Show that the anticommutator of two Pauli matrices is

$$\{\sigma_i, \sigma_j\} = 2\delta_{ij}. \quad (7.170)$$

7.5 Let $\mathbf{n}$ be any unit vector and $\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ be the vector whose components are the Pauli matrices. Why is it physically necessary that $\mathbf{n} \cdot \boldsymbol{\sigma}$ satisfy $(\mathbf{n} \cdot \boldsymbol{\sigma})^2 = I$, where I is the 2×2 identity matrix? Let $\mathbf{m}$ be a unit vector such that $\mathbf{m} \cdot \mathbf{n} = 0$. Why do we require that the commutator $[\mathbf{m} \cdot \boldsymbol{\sigma}, \mathbf{n} \cdot \boldsymbol{\sigma}] = 2i(\mathbf{m} \times \mathbf{n}) \cdot \boldsymbol{\sigma}$? Prove that these relations follow from the algebraic properties of the Pauli matrices. You should be able to show that $[\mathbf{m} \cdot \boldsymbol{\sigma}, \mathbf{n} \cdot \boldsymbol{\sigma}] = 2i(\mathbf{m} \times \mathbf{n}) \cdot \boldsymbol{\sigma}$ for any two vectors $\mathbf{n}$ and $\mathbf{m}$.

7.6 Let $\mathbf{n}$ be the unit vector in the direction with polar coordinates (θ, ϕ) . Write down the matrix $\mathbf{n} \cdot \boldsymbol{\sigma}$ and find its eigenvectors. Hence show that the state of a spin-half particle in which a measurement of the component of spin along $\mathbf{n}$ is certain to yield $\frac{1}{2}\hbar$ is

$$|+, \mathbf{n}\rangle = \sin(\theta/2)e^{i\phi/2}|- \rangle + \cos(\theta/2)e^{-i\phi/2}|+ \rangle, \quad (7.171)$$

where $|\pm\rangle$ are the states in which $\pm\frac{1}{2}\hbar$ is obtained when s_z is measured. Obtain the corresponding expression for $|- , \mathbf{n}\rangle$. Explain physically why the amplitudes in (7.171) have modulus $2^{-1/2}$ when $\theta = \pi/2$ and why one of the amplitudes vanishes when $\theta = \pi$.

7.7 For a spin-half particle at rest, the rotation operator $\mathbf{J}$ is equal to the spin operator $\mathbf{S}$. Use the result of Problem 7.4 to show that in this case the rotation operator $U(\boldsymbol{\alpha}) = \exp(-i\boldsymbol{\alpha} \cdot \mathbf{J})$ is

$$U(\boldsymbol{\alpha}) = I \cos\left(\frac{\alpha}{2}\right) - i\hat{\boldsymbol{\alpha}} \cdot \boldsymbol{\sigma} \sin\left(\frac{\alpha}{2}\right), \quad (7.172)$$

where $\hat{\boldsymbol{\alpha}}$ is the unit vector parallel to $\boldsymbol{\alpha}$. Comment on the value this gives for $U(\boldsymbol{\alpha})$ when $\alpha = 2\pi$.

7.8 Write down the 3×3 matrix that represents S_x for a spin-one system in the basis in which S_z is diagonal (i.e., the basis states are $|0\rangle$ and $|\pm\rangle$ with $S_z|+\rangle = |+\rangle$, etc.)

A beam of spin-one particles emerges from an oven and enters a Stern-Gerlach filter that passes only particles with $J_z = \hbar$. On exiting this filter, the beam enters a second filter that passes only particles with $J_x = \hbar$, and then finally it encounters a filter that passes only particles with $J_z = -\hbar$. What fraction of the particles stagger right through?

7.9* Repeat the analysis of Problem 7.8 for spin-one particles coming on filters aligned successively along $+z$, 45° from z towards x [i.e. along $(1,0,1)$], and along x .

Use classical electromagnetic theory to determine the outcome in the case that the spin-one particles were photons and the filters were Polaroid. Why do you get a different answer?

7.10 A system that has spin momentum $\sqrt{6}\hbar$ is rotated through an angle ϕ around the z -axis. Write down the 5×5 matrix that updates the amplitudes a_m that S_z will take the value m .

7.11 Justify physically the claim that the Hamiltonian of a particle that precesses in a magnetic field $\mathbf{B}$ can be written

$$H = -2\mu\mathbf{s} \cdot \mathbf{B}. \quad (7.173)$$

In a coordinate system oriented such that the z -axis is parallel to $\mathbf{B}$, a proton is initially in the eigenstate $|+, x\rangle$ of s_x . Obtain expressions for the expectation values of s_x and s_y at later times. Explain the physical content of your expressions.

Bearing in mind that a rotating magnetic field must be a source of radiation, do you expect your expressions to remain valid to arbitrarily late times? What really happens in the long run?

7.12 Show that a classical top with spin angular momentum $\mathbf{S}$ which is subject to a torque $\mathbf{G} = \mu\mathbf{S} \times \mathbf{B}/|\mathbf{S}|$ precesses at angular velocity $\boldsymbol{\omega} = \mu\mathbf{B}/|\mathbf{S}|$. Explain the relevance of this calculation to magnetic resonance imaging in general and equation (7.70b) in particular.

7.13* Write a computer program that determines the amplitudes a_m in

$$|\mathbf{n}; s, s\rangle = \sum_{m=-s}^s a_m |s, m\rangle$$

where $\mathbf{n} = (\sin\theta, 0, \cos\theta)$ with θ any angle and $|\mathbf{n}; s, s\rangle$ is the ket that solves the equation $(\mathbf{n} \cdot \mathbf{S})|\mathbf{n}; s, s\rangle = s|\mathbf{n}; s, s\rangle$. Explain physically the nature of this state.

Use your a_m to evaluate the expectation values $\langle S_x \rangle$ and $\langle S_x^2 \rangle$ for this state and hence show that the rms fluctuation in measurements of S_x will be $\sqrt{s/2} \cos\theta$.

7.14* We have that

$$L_+ = L_x + iL_y = e^{i\phi} \left(\frac{\partial}{\partial\theta} + i \cot\theta \frac{\partial}{\partial\phi} \right). \quad (7.174)$$

From the Hermitian nature of $L_z = -i\partial/\partial\phi$ we infer that derivative operators are anti-Hermitian. So using the rule $(AB)^\dagger = B^\dagger A^\dagger$ on equation (7.174), we infer that

$$L_- = L_+^\dagger = \left(-\frac{\partial}{\partial\theta} + i \frac{\partial}{\partial\phi} \cot\theta \right) e^{-i\phi}.$$

This argument and the result it leads to is wrong. Obtain the correct result by integrating by parts $\int d\theta \sin\theta \int d\phi (f^* L_+ g)$, where f and g are arbitrary functions of θ and ϕ . What is the fallacy in the given argument?

7.15* By writing $\hbar^2 L^2 = (\mathbf{x} \times \mathbf{p}) \cdot (\mathbf{x} \times \mathbf{p}) = \sum_{ijklm} \epsilon_{ijk} x_j p_k \epsilon_{ilm} x_l p_m$ show that

$$p^2 = \frac{\hbar^2 L^2}{r^2} + \frac{1}{r^2} \{(\mathbf{r} \cdot \mathbf{p})^2 - i\hbar \mathbf{r} \cdot \mathbf{p}\}. \quad (7.175)$$

By showing that $\mathbf{p} \cdot \hat{\mathbf{r}} - \hat{\mathbf{r}} \cdot \mathbf{p} = -2i\hbar/r$, obtain $\mathbf{r} \cdot \mathbf{p} = r p_r + i\hbar$. Hence obtain

$$p^2 = p_r^2 + \frac{\hbar^2 L^2}{r^2}. \quad (7.176)$$

Give a physical interpretation of one over $2m$ times this equation.

7.16 The angular part of a system's wavefunction is

$$\langle \theta, \phi | \psi \rangle \propto (\sqrt{2} \cos \theta + \sin \theta e^{-i\phi} - \sin \theta e^{i\phi}).$$

What are the possible results of measurement of (a) L^2 , and (b) L_z , and their probabilities? What is the expectation value of L_z ?

7.17 A system's wavefunction is proportional to $\sin^2 \theta e^{2i\phi}$. What are the possible results of measurements of (a) L_z and (b) L^2 ?

7.18 A system's wavefunction is proportional to $\sin^2 \theta$. What are the possible results of measurements of (a) L_z and (b) L^2 ? Give the probabilities of each possible outcome.

7.19 Consider a stationary state $|E, l\rangle$ of a free particle of mass m that has angular-momentum quantum number l . Show that $H_l |E, l\rangle = E |E, l\rangle$, where

$$H_l = \frac{1}{2m} \left(p_r^2 + \frac{l(l+1)\hbar^2}{r^2} \right). \quad (7.177)$$

Give a physical interpretation of the two terms in the big bracket. Show that $H_l = A_l^\dagger A_l$, where

$$A_l = \frac{1}{\sqrt{2m}} \left(i p_r - \frac{(l+1)\hbar}{r} \right). \quad (7.178)$$

Show that $[A_l, A_l^\dagger] = H_{l+1} - H_l$. What is the state $A_l |E, l\rangle$? Show that for $E > 0$ there is no upper bound on the angular momentum. Interpret this result physically.

7.20* Show that $[J_i, L_j] = i \sum_k \epsilon_{ijk} L_k$ and $[J_i, L^2] = 0$ by eliminating L_i using its definition $\mathbf{L} = \hbar^{-1} \mathbf{x} \times \mathbf{p}$, and then using the commutators of J_i with $\mathbf{x}$ and $\mathbf{p}$.

7.21* In this problem you show that many matrix elements of the position operator $\mathbf{x}$ vanish when states of well-defined l, m are used as basis states. These results will lead to selection rules for electric dipole radiation. First show that $[L^2, x_i] = i \sum_{jk} \epsilon_{jik} (L_j x_k + x_k L_j)$. Then show that $\mathbf{L} \cdot \mathbf{x} = 0$ and using this result derive

$$[L^2, [L^2, x_i]] = i \sum_{jk} \epsilon_{jik} (L_j [L^2, x_k] + [L^2, x_k] L_j) = 2(L^2 x_i + x_i L^2). \quad (7.179)$$

By squeezing this equation between angular-momentum eigenstates $\langle l, m |$ and $| l', m' \rangle$ show that

$$0 = \{(\beta - \beta')^2 - 2(\beta + \beta')\} \langle l, m | x_i | l', m' \rangle,$$

where $\beta \equiv l(l+1)$ and $\beta' \equiv l'(l'+1)$. By equating the factor in front of $\langle l, m | x_i | l', m' \rangle$ to zero, and treating the resulting equation as a quadratic equation for β given β' , show that $\langle l, m | x_i | l', m' \rangle$ must vanish unless $l+l' = 0$ or $l = l' \pm 1$. Explain why the matrix element must also vanish when $l = l' = 0$.

7.22* Show that l excitations can be divided amongst the x , y or z oscillators of a three-dimensional harmonic oscillator in $(\frac{1}{2}l+1)(l+1)$ ways. Verify in the case $l = 4$ that this agrees with the number of states of well-defined angular momentum and the given energy.

7.23* Let

$$A_l = \frac{1}{\sqrt{2m\hbar\omega}} \left(ip_r - \frac{(l+1)\hbar}{r} + m\omega r \right). \quad (7.180)$$

be the ladder operator of the three-dimensional harmonic oscillator and $|E, l\rangle$ be the stationary state of the oscillator that has energy E and angular-momentum quantum number l . Show that if we write $A_l|E, l\rangle = \alpha_-|E - \hbar\omega, l+1\rangle$, then $\alpha_- = \sqrt{\mathcal{L}-l}$, where $\mathcal{L}$ is the angular-momentum quantum number of a circular orbit of energy E . Show similarly that if $A_{l-1}^\dagger|E, l\rangle = \alpha_+|E + \hbar\omega, l-1\rangle$, then $\alpha_+ = \sqrt{\mathcal{L}-l+2}$.

7.24* Show that the probability distribution in radius of a particle that orbits in the three-dimensional harmonic oscillator potential on a circular orbit with angular-momentum quantum number l peaks at $r/\ell = \sqrt{2(l+1)}$, where

$$\ell = \sqrt{\frac{\hbar}{2m\omega}}. \quad (7.181)$$

Derive the corresponding classical result.

7.25* A particle moves in the three-dimensional harmonic oscillator potential with the second largest angular-momentum quantum number possible at its energy. Show that the radial wavefunction is

$$u_1 \propto x^l \left(x - \frac{2l+1}{x} \right) e^{-x^2/4} \quad \text{where } x = r/\ell \quad \text{with } \ell = \sqrt{\frac{\hbar}{2m\omega}}. \quad (7.182)$$

How many radial nodes does this wavefunction have?

7.26 A box containing two spin-one gyros A and B is found to have angular-momentum quantum numbers $j = 2$, $m = 1$. Determine the probabilities that when J_z is measured for gyro A, the values $m = \pm 1$ and 0 will be obtained.

What is the value of the Clebsch-Gordan coefficient $C(2, 1; 1, 1, 1, 0)$?

7.27 The angular momentum of a hydrogen atom in its ground state is entirely due to the spins of the electron and proton. The atom is in the state $|1, 0\rangle$ in which it has one unit of angular momentum but none of it is parallel to the z -axis. Express this state as a linear combination of products of the spin states $|\pm, e\rangle$ and $|\pm, p\rangle$ of the proton and electron. Show that the states $|x\pm, e\rangle$ in which the electron has well-defined spin along the x -axis are

$$|x\pm, e\rangle = \frac{1}{\sqrt{2}} (|+, e\rangle \pm |-, e\rangle). \quad (7.183)$$

By writing

$$|1, 0\rangle = |x+, e\rangle\langle x+, e|1, 0\rangle + |x-, e\rangle\langle x-, e|1, 0\rangle, \quad (7.184)$$

express $|1, 0\rangle$ as a linear combination of the products $|x\pm, e\rangle|x\pm, p\rangle$. Explain the physical significance of your result.

7.28* The interaction between neighbouring spin-half atoms in a crystal is described by the Hamiltonian

$$H = K \left(\frac{\mathbf{S}^{(1)} \cdot \mathbf{S}^{(2)}}{a} - 3 \frac{(\mathbf{S}^{(1)} \cdot \mathbf{a})(\mathbf{S}^{(2)} \cdot \mathbf{a})}{a^3} \right), \quad (7.185)$$

where K is a constant, $\mathbf{a}$ is the separation of the atoms and $\mathbf{S}^{(1)}$ is the first atom's spin operator. Explain what physical idea underlies this form of H . Show that $S_x^{(1)} S_x^{(2)} + S_y^{(1)} S_y^{(2)} = \frac{1}{2} (S_+^{(1)} S_-^{(2)} + S_-^{(1)} S_+^{(2)})$. Show that the mutual eigenkets of the total spin operators S^2 and S_z are also eigenstates of H and find the corresponding eigenvalues.

At time $t = 0$ particle 1 has its spin parallel to $\mathbf{a}$, while the other particle's spin is antiparallel to $\mathbf{a}$. Find the time required for both spins to reverse their orientations.